The Shadow Cost of Credit: Payment Choice and Credit Capacity*

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Abstract

Linking 1.5 million delivery orders to households' daily credit positions, we find that a household's cash use rises in a convex pattern with its credit utilization. We explain this pattern with a model in which households preserve available credit for unexpected expenses that require credit. As utilization rises, the remaining credit is more likely to fall short of such an expense; therefore, each dollar of routine spending paid on credit carries a higher shadow cost, and households substitute cash for credit. Using a regression discontinuity design, we show that bank-initiated limit increases reduce cash use, most among households near their limit, as the model predicts. We estimate that for a household that pays no interest, a dollar on credit carries a shadow cost of 5 cents at 20 percent utilization and 24 cents at 80 percent, close to the 25 cents of cumulative interest on a dollar of revolving debt.

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1 Introduction

In the United States, credit cardholders who carry no revolving debt and earn rewards pay with credit for only 65.5 percent of in-person purchases (Diary of Consumer Payment Choice, 2025). If cash and credit were fungible, these households would appear to leave money on the table one third of the time. This paper argues that they pay in cash today to preserve available credit for unanticipated expenses that require it tomorrow, such as hotel pre-authorizations and hospital deposits. Each dollar charged now increases the likelihood that the remaining credit line will fall short of such an expense. Therefore, every purchase on credit carries a shadow cost even when no interest accrues, and this cost grows as available credit shrinks. This shadow cost captures the insurance value of credit capacity. Firms likewise hold cash and unused credit against different risks (Acharya et al., 2007; Lins et al., 2010), but how much households value unused credit beyond cash liquidity has not been measured.

This paper measures the shadow cost of credit from how a household's choice between cash and credit responds to its credit capacity. On Rappi, an on-demand delivery platform in Mexico, the price is the same whether a household pays in cash or by card, so its choice reflects the cost of each instrument as the household perceives it. We observe the same household's choices for similar goods over time, each linked to its credit position on the day before. Controlling for household fixed effects, a household's cash use rises in a convex pattern as credit utilization grows. We build a model of payment choice in which a rising cost of credit generates this pattern. As the model predicts, bank-initiated limit increases reduce cash use, most among households with high utilization. Matching the model to the cash shares, we estimate that for a household that pays no interest, the shadow cost of a dollar on credit rises from 5 cents at 20 percent utilization to 24 cents at 80 percent, nearly the 25 cents of cumulative interest on a dollar of revolving debt.

We link Rappi's orders to the records of cardholders of RappiCard, a general-purpose credit card integrated with Rappi, between 2021 and 2025. The card's transaction ledger records each household's daily credit utilization and interest-bearing balances. Detailed consumption data from Rappi provide information on items, prices, timing, and payment

methods for every order. For households with recent credit bureau data, the data also include limits and balances on other credit cards, enabling us to measure their aggregate monthly credit utilization. To the best of our knowledge, this is the first dataset to record each transaction's payment method alongside the household's daily credit position. The sample includes 73,000 households and 1.5 million orders in the first year after card approval. Cash pays 9.5 percent of the orders, and over two-fifths of households use cash at least once. Also, the median cash order occurs within seven days of a card order by the same household. Thus, payment choice varies within the household over short spans.

Within a household, we identify two distinct patterns in the relationship between cash payments and the household's credit position. First, cash usage rises convexly with RappiCard credit utilization, regardless of whether the household pays interest, controlling for basket characteristics and time fixed effects. Second, cash payments are more frequent when a household carries interest-bearing debt, at every utilization level. Both patterns hold when RappiCard utilization is replaced with the household's aggregate credit position across all its cards. A simple comparison of the marginal cost of cash and credit does not account for either pattern: cashback favors credit, and the interest cost of revolving debt remains constant with utilization. Credit line size, credit-score management, cash on hand, and salience of utilization also do not generate them.

To explain these patterns, we build a three-period model of payment choice. A household chooses between cash and credit payments for a fixed-size order by comparing costs, while anticipating future expenditure shocks that require credit. If available credit is insufficient for such shocks, the household incurs a higher cost to finance the excess, such as informal credit or paying late at a penalty, creating an incentive to preserve credit capacity. The cost of credit has two parts: the interest a dollar of credit accrues, which only revolving households pay, and a shadow cost, the expected cost of financing the excess should a shock exceed available credit, which every household bears. Each dollar charged raises the chance of an overflow. Because the density of the expenditure shock is monotonically decreasing, as the data confirm, a marginal dollar of credit increases overflow risk more as a household approaches its limit. As a result, the cost of credit is convex and increases with utilization, and it is always higher when a household is revolving. An order is paid with cash when the

cost of cash falls below that of credit, so the model explains both empirical patterns.

The model yields several testable predictions: a credit limit increase at an unchanged balance lowers the shadow cost and leaves the interest cost unchanged, so cash use should fall, most among high-utilization households, and by a similar amount for revolving and non-revolving households. We test these predictions using credit limit increase (CLI) campaigns initiated by the card issuer in a fuzzy regression discontinuity design. A higher credit limit lowers the number of cash-paid orders in the following month. The decline is concentrated among households with utilization above 60 percent, where the probability of placing any cash order falls by 4.5 percentage points against 10.5 percent just below the cutoff. Moreover, revolving and non-revolving households reduce cash-paid orders by similar amounts. A voluntary repayment, by contrast, reduces cash use more for revolving households near the limit than for non-revolving households.

The final section of the paper estimates the model and assigns monetary values to the two components of the credit cost. In the model, at a given credit position, the cash share is the fraction of payments whose cost of credit exceeds their cost of cash; thus, matching observed cash shares across credit positions recovers the cost of credit and the distribution of the cost of cash. We measure the interest part from the ledger of revolved balances, as the cumulative interest on a dollar of revolving debt until it is repaid or payments stop, and the shadow cost is what remains. For a household that pays no interest, the shadow cost increases from 5 cents per dollar at 20 percent utilization to 24 cents at 80 percent utilization, close to the 25 cents of cumulative interest on a dollar of revolving balance. This explains why households that pay no interest switch to cash as utilization rises: the cost of credit depends on utilization as well as revolving status, and near the limit the shadow cost alone matches the interest cost that revolving households pay.

Finally, we recompute the cost of credit a household faces after a 10 percent cut in the rate and a 10 percent increase in the limit, holding its spending and repayment fixed, and show that the price and the quantity of credit move different components of the cost. The rate cut lowers only the interest component, so it reaches revolving households alone and by a similar amount at every utilization. In contrast, the limit increase expands the insurance that available credit provides: it lowers the shadow cost for every household, and

the reduction is largest for those near their limit. The limit increase also shifts purchases from cash to credit in the model, so part of the rise in card balances after a limit increase documented in the literature reflects existing purchases rather than new borrowing.

This paper contributes to three strands of literature. The first examines household responses to changes in credit lines. Cardholders borrow more when their credit limit rises, and the response is largest for those near the limit (Gross and Souleles, 2002; Agarwal et al., 2018; Aydin, 2022; Bord et al., 2025). The response below the limit is read as precautionary (Gross and Souleles, 2002): a limit that does not bind today may bind tomorrow.[1] A separate set of findings using consumer surveys suggests revolvers and cardholders near their limits use debit more often (Zinman, 2009; Sprenger and Stavins, 2008; Stavins, 2020). Although researchers have identified the precautionary motive and payment shift separately, they have not linked them or treated them together as a cost of credit. This paper observes the same shift to cash within the household as utilization rises, identifies it causally from a bank-initiated limit increase, and prices the motive for preserving available credit behind it.

Second, the paper speaks to the credit card debt puzzle: households carry costly revolving debt while holding low-yield liquid assets. Three explanations exist for this phenomenon. The first explanation is a restriction in transaction technology, where certain obligations necessitate cash payments, prompting households to retain liquidity alongside credit card debt (Telyukova and Wright, 2008; Telyukova, 2013; Greene and Stavins, 2023). The second explanation is credit access risk, where households preemptively borrow and retain liquid savings against the risk of credit limit reductions (Fulford, 2015; Druedahl and Jørgensen, 2018; Gorbachev and Luengo-Prado, 2019). The third explanation involves mental accounting. Using daily transaction data, Gathergood and Olafsson (2024) show that co-holding results from households assigning specific spending categories to overdraft accounts.[2] All

[1]Related work studies the growth of limits and balances over the life cycle (Fulford and Schuh, 2024), interest rates and minimum payments in the Mexican market with default as the outcome (Castellanos et al., 2025), lender-initiated limit reductions (Chava et al., 2023), debt relief (Dobbie and Song, 2020), and minimum payment rules in the United States (Keys and Wang, 2019) and Mexico (Medina and Negrín, 2022).

[2]Behavioral and self-control explanations for co-holding include the accountant-shopper and partner-control models (Bertaut et al., 2009; Medina and Pagel, 2025; Vihriälä, 2025), mental accounting of savings and debt (Prelec and Loewenstein, 1998), repayment heuristics (Gathergood et al., 2019), and a preference for debit over credit spending (Batista et al., 2025).

three explanations account for the stock level of debt and liquid assets on a household's balance sheet. This paper adds a shadow cost of credit to the payment decision. Households keep available credit for expenses that require credit, and cash for expenses that require cash; together, these two needs can accumulate into co-holding.

Third, this study speaks to the literature on payment choice. Shy (2023) provides a comprehensive review of this literature. The canonical model conceptualizes the cash versus credit decision as a cash inventory problem. Holding cash is costly, so households spend the cash they have before using a credit card (Alvarez and Lippi, 2017), and structural estimates of this model evaluate the cost of the card against the cost of cash (Briglevics and Schuh, 2020; Lippi and Moracci, 2026).[3] The literature further explores how the non-fungibility between instruments paying for shocks generates precautionary demand for cash (Alvarez and Lippi, 2013) and demand for cash as a backup when digital payment systems fail (Alvarez et al., 2026). In Mexico, most Uber riders possess both a registered card and cash and alternate between them (Alvarez and Argente, 2025). This paper adds credit position as a state variable in payment choice: available credit is the inventory for expenses that require credit, and cash has the value of preserving this inventory.

The rest of the paper proceeds in this order: setting and data (Section 2), key empirical patterns (Section 3), conceptual model (Section 4), causal inference and testing model predictions (Section 5), estimation and counterfactual (Section 6), and conclusion (Section 7).

2 Setting and Data

2.1 RappiCard and Rappi

Cash remains the dominant means of payment in Mexico: even cardholders paid about 90 percent of their transactions with cash (Alvarez et al., 2022). For a household with credit available at the point of sale, the choice between credit and cash arises at nearly every purchase. We observe this choice order by order on Rappi among holders of RappiCard, a

[3] In these models, the cost of cash withdrawal determines the cash stock level (Alvarez and Lippi, 2009). In the new monetarist theory, the debt limit arises from limited commitment, necessitating the use of money where credit is impractical (Wang et al., 2017). Within the structural adoption-and-use framework, the cost of each instrument is a household-specific parameter (Koulayev et al., 2016).

credit card integrated into the platform.

RappiCard Mexico is a general-purpose credit card launched in January 2021 as a joint venture between Banorte, one of Mexico's largest banks, and Rappi, the leading on-demand delivery platform in Latin America.[4] The contract follows the standard format of Mexican bank cards. Statements are posted every 30 days, and payment is due about 20 days later, so a purchase paid in full gets up to 50 days of interest-free financing, while a cardholder who pays less than the statement balance pays interest on the balance carried (Chioda et al., 2026). RappiCard charges no annual fee and has an average annual percentage rate (APR) of 72 percent. This APR is above the rate of traditional banks (HSBC Zero, 46 percent) and below that of fintech issuers (Nu Bank, 95 percent).[5] Lastly, RappiCard offers 3 percent cashback on Rappi purchases, increasing up to 5 percent for Rappi Pro subscribers, and 1 percent at other merchants. The card is accepted at point-of-sale terminals and online merchants.

Rappi delivers restaurant, grocery, pharmacy, and convenience store orders to the customer's door. Figure 1 shows two screens from the app. Panel A shows that the RappiCard account is accessible from the Rappi app's home page, allowing users to check their balance and remaining credit with a single tap. At checkout, the cardholder selects a payment method from the list shown in Panel B before confirming the order: a linked bank card, a third-party wallet such as Apple Pay, or cash on delivery. The checkout page displays no available credit for RappiCard or for any other card, so no card's utilization is made more salient. For cash orders, the courier collects payment and provides change at delivery.

[Figure 1 about here.]

Each Rappi order can be matched to the credit card records of RappiCard holders at the time it was placed because Rappi constantly shares its user activity data with RappiCard. Applications to RappiCard are made online and require an existing Rappi account. RappiCard uses the applicant's purchase history on Rappi to make approval decisions and continues to receive cardholders' Rappi activity post-approval.

[4] On the launch and ownership of RappiCard, see Mexico Business News, December 2025.

[5] Weighted average annual interest rates disclosed by each issuer in 2026: RappiCard, HSBC Zero, and Nu.

2.2 Data

RappiCard Mexico provided a random sample of about 99,000 cardholders approved between January 2021 and November 2025. We build an order-level estimation sample that links a household's daily RappiCard balances, revolving status, and monthly utilization on other credit cards to Rappi order details.[6] This is a rare setting where a household's payment choice and credit position are observed simultaneously from granular consumption data.

Rappi orders The Rappi order data arrive at the item level, with each item's price and product description. We use the product descriptions to classify each item as an essential, a temptation good (alcohol or tobacco), or other, then aggregate to the order level. Each order records the payment method, order time, total value, number of items, platform discount, store, and category.

The payment choice on Rappi reflects the household's perceived cost of the two instruments because the platform removes the usual confounds of instrument choice. At checkout, the customer chooses among linked bank cards, third-party wallets, and cash at the same price: item and order values are confirmed before the payment method is selected. Every order is recorded regardless of the payment method, so cash gives the merchant no tax or informality advantage.

RappiCard records RappiCard provided several data sets at different levels. First, the transaction ledger includes every spending and repayment transaction, settled or not. We use it to construct the cardholder's balance on the day before each order and its monthly spending by category. Second, the billing statement file records the statement balance, revolving balance, interest and fees charged, outstanding installment balances, and whether the account was blocked for non-payment at the billing-cycle level. Lastly, we observe households' fixed characteristics at approval, including gender, age, and zip code, from the application data.

[6] We refer to each cardholder as a household; credit positions, including utilization on other cards, are measured for the cardholder.

Credit bureau records RappiCard pulls each cardholder's credit bureau file about every two months. Each snapshot shows a household's bureau score, the length of the credit history, and, for every tradeline, the limit, balance, payment, and delinquency status. These records inform the household's overall credit condition beyond RappiCard.

Credit limit increase campaigns A separate data set covers the credit limit increase campaigns RappiCard ran in 2023 and 2024. For each campaign, it records every cardholder RappiCard evaluated, the bureau credit score, its limit and utilization at evaluation, whether it was offered an increase, the size of the increase, and whether it accepted the increase. Matching these records to the main sample by user gives a user-by-campaign panel of 38,640 users over four campaigns. Cash-paid orders, spending, and revolving status in the weeks after each campaign come from the order and statement records and are merged onto this panel.

Census and cash access We link each household to census data through its home location, approximated by the median coordinates of its app launches after 10 p.m., or by the zip code at application for users who never launch at night. Each address is mapped to an AGEB, the basic census unit of about 4,000 residents, and inherits its CONAPO marginalization index, a measure of local deprivation built from census income, education, and informality. In addition, using the National Directory of Economic Establishments (DENUE), we measure the distance from each home to the nearest bank branch and OXXO convenience store, Mexico's main cash access establishment.

2.3 Sample construction and measurement

The regression sample is at the order level. We match each order to the most recent statement closed before the order date, which provides the household's revolving status and prior-cycle spending, while utilization comes from the ledger on the day before the order. Orders that cannot be assigned a statement are dropped, as are months in which the account is blocked for non-payment. The estimation sample covers Rappi orders placed up to 12 months after card approval. It contains 72,632 cardholders with at least one matched order in that window

and 1.48 million orders in total. Rappi orders make up 12.5 percent of RappiCard spending, and in nearly half of the statement months with purchases, the cardholder placed no Rappi order at all. Therefore, we observe payment choice for a slice of spending on RappiCard, and a household's credit position is set mostly by spending elsewhere.

Utilization The main regressor is RappiCard utilization on the day before each order: the statement balance plus the net activity posted since the statement, divided by the prevailing limit.[7] Utilization above 100 percent is genuine. RappiCard charges no over-limit fee and routinely authorizes purchases that exceed the limit, especially for Rappi orders, which settle at almost the same rate as orders below the limit.[8] Cardholders appear to know this, and a third of the orders placed while the RappiCard is over its limit still go on the card.

Aggregate utilization For households with a recent credit bureau pull, we also construct their aggregate utilization across all credit cards at the billing-month level, as the sum of balances over the sum of limits. The RappiCard enters at its billing-month utilization and the other cards at the most recent pull, at most one month old. Households that report no other card at application and show no other credit card in the bureau have their RappiCard utilization as the aggregate. In regressions that use aggregate utilization as the explanatory variable, we drop those above 105 percent, as we do for the RappiCard measure in Section 3.

Revolving status A household is in revolving status in a billing month if the account carries an interest-bearing revolving balance or an interest-bearing installment balance in that month. Revolving status separates months in which using credit costs interest from months in which it does not.

[7] At each statement date the ledger shows the account balance; for any later day we add the purchases, payments, fees, and adjustments that have posted since that statement and divide by the prevailing limit. We exclude interest posted at the end of the billing month, so the measure is utilization before this month's interest accrues. Where the ledger shows no activity in the 30 days before an order, we use the prior statement's utilization.

[8] 95 percent of Rappi orders placed at or above the limit settle, compared with 96.7 percent of orders placed below it, computed on Rappi purchases in the first 12 months after card approval with utilization merged from the transaction ledger.

2.4 Summary statistics and stylized facts

Summary statistics Table 1 describes the estimation sample at user, user-month and order levels. First, RappiCard holders are young but experienced card users in a country where only 33 percent of adults hold any formal credit (INEGI and CNBV, 2021). Over 90 percent of them held another credit card at approval, and RappiCard accounts for about a quarter of a median household's total credit card limit. Their median age is 28.5 years, and half have 7.5 years or less of credit history.

Second, households use the card heavily, and utilization varies widely. Across user-months, the balance averages 56.3 percent of the limit, with an interquartile range from 18 percent to 95 percent. Average aggregate utilization is about the same where we can measure it. Households are in revolving status in 45 percent of billing months, even in their first year with RappiCard. In these revolving months, the median interest-bearing balance nearly matches the median credit line at RappiCard.

A Rappi order is small relative to the credit line. Panel C shows that the median order is 208 pesos (about USD 12), about 4 percent of the median initial limit, so for most orders the limit does not force the household into cash. Most orders go on a card: 59.2 percent on the RappiCard and 28.3 percent on another card, while 9.5 percent are paid in cash.

[Table 1 about here.]

Cash prevalence First, although cash is a minority instrument on the platform, cardholders use it extensively. Within the year, 42.6 percent of users pay cash at least once, and among them the median share of orders paid in cash is 25 percent. Moreover, users switch payment methods over short periods. The median cash-paid order is placed within seven days of a card-paid order by the same user. Therefore, cash payments are not limited to a distinct population or specific time periods.

Second, cash is used even when paying on credit is free. Half of cash-paid orders, 49.6 percent, are placed in months with no interest due, when charging the order would have earned cashback at no cost. The other half fall in revolving months, where cash pays 11.8 percent of orders.

Third, the credit state shifts cash use across purchase types. Appendix Table D1 reports the share of cash-paid orders by category and revolving state. In non-revolving months, restaurant orders are paid in cash 8.3 percent of the time, more often than grocery and temptation goods. Within each category, orders placed in revolving months are paid in cash 3 to 5 percentage points more often.

Fourth, cash orders are placed at higher utilization than card orders in both revolving states. In non-revolving months, the median RappiCard utilization on the day before a cash-paid order is 46 percent, 15 percentage points above that of the median card-paid order. Revolving households have higher utilization on average, but their cash use follows a similar pattern.

What predicts cash payment We ask which characteristics of an order matter most for whether it is paid in cash. We estimate a logit of the cash indicator on five blocks of covariates: credit position (day-before utilization and revolving status), purchase category, timing (payday proximity, time of day, and position in the billing cycle), prior-cycle card spending and repayment, and order size, with calendar-month dummies and the household's average cash rate to absorb platform-wide trends and each household's baseline propensity to use cash. To measure out-of-sample power, we split households into five groups, fit the model on four, and predict the fifth, so no order is predicted from its own household's coefficients. We then drop one block at a time and record how much the model's predictive power falls, measured by area under the curve (AUC).

Pooled over all orders, the model separates cash from card orders with an AUC of 0.612: if we draw one cash-paid order and one card-paid order at random, the model assigns the higher cash probability to the cash-paid order 61.2 percent of the time, where a coin flip scores 50 percent. Figure 2 plots the drop in AUC when each block is removed. Credit position is the most important predictor: removing it costs 3.7 percentage points, a third of the model's advantage over chance and twice the cost of the next block, prior-cycle spending and repayment. No block beyond those two costs more than half a point.

[Figure 2 about here.]

Cash pays a minority of orders, but it is spread across many households that also pay by card, half of it in months when credit is free. Among the features of an order, the credit position is the strongest predictor of which orders go to cash. Section 3 holds the household fixed and traces its cash use against its credit position.

3 The Cash-Credit Gradient

3.1 The within-person gradient

We investigate the shape of the relationship between cash payment and credit utilization while controlling for other blocks of predictors. We sort orders into five bins of the household's utilization on the day before the order, and then run the regression at the order level,

$$\begin{aligned} \text{cash}_{io} \;=\; & \sum_{k=2}^{5} \beta_k \, \mathbf{1}\{u_{io} \in B_k\} \;+\; \gamma \, R_{im} \;+\; \sum_{k=2}^{5} \delta_k \, \mathbf{1}\{u_{io} \in B_k\} \times R_{im} \\ & +\; X_{io}'\lambda \;+\; \alpha_i + \tau_m + \mu_c + \phi_d + \varepsilon_{io}. \end{aligned} \tag{1}$$

where cash_{io} equals one if household i pays order o in cash. u_{io} is its day-before utilization, and B_2 through B_5 are indicators that utilization falls in $[20\%, 40\%)$, $[40\%, 60\%)$, $[60\%, 80\%)$, and $[80\%, +)$, with utilization below 20 percent omitted. The coefficients β_k trace the cash response to a household's utilization in non-revolving months. R_{im} equals one if the account carries an interest-bearing balance in the billing month prior to the order, and its coefficient measures the difference in cash-use probability between revolving and non-revolving months in the omitted bin. The coefficients on the interactions between bin indicators and the revolving state, δ_k, indicate the extra gap in bin k. The regression controls for the natural log of order value and RappiCard spending in previous months. In addition, for the subset of households with a recent credit bureau pull, we add a range of controls to absorb a common distress channel, in which a household in trouble on another line runs up its RappiCard and pays more cash for the same underlying reason.[9]

[9]The control variables built from credit bureau files include: the natural log of limits on their other tradelines, the number of active cards, whether the household has a mortgage, a bank tradeline, or a microfinance tradeline, and whether any other tradeline is delinquent or charged off.

We add a user fixed effect α_i and a billing-cycle fixed effect τ_m to the specification, so the coefficients always compare orders of the same value made by a household across months with similar consumption patterns. Two further fixed effects refine the comparison. The category effect μ_c absorbs differences in what is bought. The cycle-position effect ϕ_d absorbs differences based on where an order falls in the billing cycle.[10] We control for cycle position because a statement just issued or a payment coming due makes the credit position salient in a way that also varies with utilization, so this effect compares orders at similar points in the cycle. The predictive exercise in Figure 2 uses the same split. Standard errors are clustered by user. The estimation sample includes orders with day-before utilization at or below 105 percent and a positive value: 1.28 million orders from 70,000 households.

Table 2 reports the estimates. In non-revolving months, when interest is not charged on purchases, cash use rises with utilization: the coefficients rise with utilization and are significant from 40 percent on. This increase in cash use is convex — gentle through the middle bins, but steep in the highest ones. Across the utilization spectrum, a household in a revolving month consistently pays with cash more often than when it is not, and this gap grows as utilization approaches the limit. Adding order controls and category and cycle-position fixed effects barely moves the coefficients (columns 1 to 3), and in the subsample with credit bureau controls (column 4), cash use in each revolving state follows nearly the same path as in column 3. In the preferred specification in column 3, a household pays cash 1.9 percentage points more often when it is in revolving status and has used at least 80 percent of its credit line than in a non-revolving month at the same utilization, more than a fifth of the base cash share.

[Figure 3 and Table 2 about here.]

Figure 3 turns the column-3 estimates into levels. Column 3 is our preferred specification with the full range of controls and fixed effects on the full sample. The figure plots the fitted

[10]This is a five-way split into the three days after the statement, mid-cycle, the five days before the payment due date, the five days after it, and the run-up to the next statement. The two groups bracketing the payment due date are defined by the distance between the order date and the due date of its cycle and take precedence over the cycle-day groups, so an order within five days on either side of the due date is assigned to that window even if it would otherwise fall mid-cycle. Orders in a cycle with no purchase, and hence no due date, remain in the mid-cycle or run-up group by days since the statement.

cash share at the bin midpoints for revolving months and non-revolving months separately, with 95 percent confidence intervals around the fitted values. Cash use climbs from roughly 7 to 10 percent for a non-revolving household and from 8 to 12 percent for a revolving household as utilization increases toward the limit. Declined card payments cannot drive this rise: RappiCard authorizes over-limit Rappi orders over 95 percent of the time (Section 2), and in finer bins cash use climbs steadily from 80 to 100 percent, before any order exceeds the limit (Appendix Figure C1).

Cash payment and aggregate credit capacity If the household values available credit capacity as a buffer, its cash use should respond to the aggregate utilization across all credit cards. Since RappiCard accounts for part of the credit limit for the majority of our sample households, cash response should stay flat over most of the range of aggregate utilization and rise only as total capacity nearly depletes. We re-estimate equation (1) with the bins defined on aggregate utilization from Section 2 and revolving status based on RappiCard. Table 3 reports the estimates. Below 60 percent aggregate utilization, non-revolving households do not use cash significantly more as usage grows, but cash use increases by 1.8 percentage points in the top utilization bin, about a fifth of the base cash share. The higher tendency to pay cash at any utilization level when a household is revolving also stays.

In addition, we expand the revolving indicator to reflect the household's aggregate credit card portfolio, proxied by the ratio of payments to balances on other credit cards.[11] Re-estimating the aggregate utilization regressions with this indicator, Appendix Table D2 shows that the aggregate gradient coefficients are essentially unchanged.

[Table 3 about here.]

Amplification across cards If cash response on Rappi is anchored only to RappiCard utilization, the other cards' state should not affect the cash response to it. We test this by interacting RappiCard utilization with the utilization of the household's other cards, each

[11]The bureau records each card's balance and payment but not whether the balance is interest-bearing, so we classify a tradeline as revolving when its payment is less than half its outstanding balance. A household is an any-card revolver if any of its other cards meet this test, and a clean non-revolver if some card is paid substantially in full and none is revolving; households whose cards fall in between are left unclassified.

in three bins, on the subsample of households for whom we observe other card balances in the credit bureau data. Figure 4 plots the estimates from Appendix Table D3. Each line holds the other cards at one utilization level and traces the cash response to RappiCard as utilization rises from below 40 percent to above 80 percent.

While RappiCard still has plenty of available credit, other cards' credit capacity hardly changes the cash response to RappiCard's utilization. However, once the RappiCard is nearly full, the effect of other cards' utilization becomes pronounced. If a household has plenty of slack on other cards, cash use rises by 2 percentage points when RappiCard utilization exceeds 80 percent. In contrast, when other cards are also nearly full, cash use rises by 3.3 points, about two thirds more. Thus, households' willingness to use cash depends on the remaining capacity across all their credit lines.

[Figure 4 about here.]

Households substitute to other cards as RappiCard credit utilization increases. Appendix Table D4 presents estimates from equation (1) for each payment instrument separately. When RappiCard utilization exceeds 80 percent, the probability that an order goes on the RappiCard falls by 8.5 percentage points. About two thirds of these orders go to other cards, and roughly 30 percent to cash.

3.2 Alternative explanations

The gradient has two features that any explanation must produce at once: cash use climbs convexly with utilization among households that pay no interest, and revolving households pay cash more often at every utilization level. We consider several alternatives to the cost-of-credit mechanism.

Credit-score management Households might pay cash to keep their reported credit utilization low and protect their credit scores. To test this, we regress credit scores on statement utilization, controlling for user and month fixed effects. Results in Appendix Table D5 show only a modest, linear effect: moving from zero to full utilization on RappiCard lowers the score by 12 points, less than a within-household standard deviation of 17 points.

There is a minor step at 30 percent utilization, the share of the limit that Mexico's financial consumer protection agency recommends staying below,[12] yet statement utilization shows no bunching at this level. Notably, while the credit score penalty on utilization is linear and limited, cash use rises much more sharply at high utilization. In addition, the bureau does not observe whether a balance bears interest, so it cannot explain why a household in revolving status uses more cash than when it is not revolving. These findings suggest that the credit score penalty on utilization is unlikely to account for the observed cash patterns. Furthermore, the credit bureau does not disclose how heavily it weights utilization in its credit-score formula, so households likely lack a clear understanding of how many points a higher balance costs them.

Credit line size Households with small credit lines tend to have high utilization. Consequently, when averaged across households, the gradient may reflect these households' characteristics rather than their credit status. We categorize users into terciles by their initial RappiCard credit limit and estimate equation (1) within each. Figure 5 shows that users with lower credit limits indeed pay cash more frequently; however, the convex increase is evident across terciles of credit limits. Additional cross-sectional evidence supports this observation: Appendix Table D6 reveals that wealthier neighborhoods, as determined by the CONAPO marginalization index introduced in Section 2, also exhibit lower average cash payments, and the convex climb appears in every marginalization tercile, though it is steeper in the poorest. Thus, the observed patterns cannot result from the composition of households.

[Figure 5 about here.]

Cash on hand A third alternative posits that the payment choice could instead depend on cash balances, as suggested in cash-inventory models (Alvarez and Lippi, 2017; Lippi and Moracci, 2026). For this mechanism to generate a gradient, cash on hand would need to increase with utilization, whereas households nearing their credit limits are often short of cash. We test the hypothesis using the payday window as a proxy for a mechanical increase in cash stocks. In Mexico, paydays occur on the 15th and the last day of the month, with

[12]The Condusef recommendation, as reported by El Siglo de Torreón, August 2026.

adjustments made to the preceding working day if these dates fall on a weekend.[13] If cash on hand determines payment choice, then replenishing cash after payday should increase cash use at all utilization levels or flatten the gradient. However, Table 4 shows that neither cash use nor its responsiveness to utilization increases after payday. In addition, a cross-person test using the variation in distance to cash access establishments supports a similar story: households living farther from a cash withdrawal point do not pay cash less often, and the gradient is slightly steeper where cash is harder to get (Appendix Table D6).[14] Thus, cash liquidity is unlikely to be the main state that drives variation in payment choice on Rappi.

[Table 4 about here.]

Salience The sharp increase in cash usage at higher utilization could also be due to a utilization alert that prompts households to switch to cash. In January 2023, RappiCard began sending a push notification urging immediate payment to every user whose utilization exceeds 70 percent, repeated after five days if the account remains above 70 percent. Appendix Table D7 re-estimates equation (1) with bins around 70 percent utilization and interacts the bins with an indicator for orders placed from January 2023 on. The results show that the convex pattern was already present before the alert. In addition, the cash response at around 70 percent utilization is slightly amplified, while the coefficient for utilization above 80 percent does not change because of the alert. Therefore, the alert alone cannot produce the observed patterns. Nevertheless, households becoming more aware of their credit position and shifting to cash payments is consistent with households valuing their available credit.

Mental accounting and temptation Households may choose cash by purchase category rather than by credit position, either because they pay for certain goods in cash based on the mental account they associate them with, or because cash serves as a commitment device against temptation (Gathergood and Olafsson, 2024; Prelec and Loewenstein, 1998). First,

[13]The Federal Labor Law caps the pay period at fifteen days (Article 88) and requires payment on a working day (Article 109).

[14]The pattern is the same when we measure cash access by the distance to the nearer of an OXXO store and a bank branch.

Figure 2 plots the predictive power of each block of explanatory variables and shows that removing purchase categories lowers the AUC by less than half a percentage point. Next, we include category fixed effects in the regressions, but they do not alter the within-person gradient — indicating that changes in cash usage are not explained by shifting types of goods purchased as utilization rises. Finally, Appendix Table D8 tests the temptation hypothesis more directly: orders containing alcohol or tobacco are not paid in cash more often than other orders, and revolving households pay cash for temptation goods only slightly less often. Together, these findings suggest that neither mental accounting nor temptation for specific goods can explain the observed patterns in cash use.

In conclusion, the within-person regressions show that remaining available credit is a key determinant of payment choice and that its effect is convex: the value the household places on its credit line rises as the line fills. The mere cost of using different payment tenders cannot explain this, nor can the alternative explanations tested above. The next section develops a conceptual model to identify a structure of the cost of credit that can jointly generate these features: a convex increase borne by all households, an additional premium specific to revolving households, and a cash response to the household's overall credit position.

4 A Conceptual Model of the Cost of Credit

The previous section shows that cash use rises in a convex pattern as the balance nears the credit limit, and that the rise is present among households that pay no interest. These patterns point to a shadow cost of using credit that increases as available credit runs out. This section builds a three-period model of payment choice, in which a household chooses how to pay for a fixed-size purchase while expecting a future expenditure shock that falls on credit. The risk of falling short of credit for such a shock creates a motive to preserve credit capacity by paying cash for the current purchase.

The model builds on the payment-technology restrictions of Telyukova and Wright (2008) and Telyukova (2013), in which the obligation itself, not the household, determines that credit and cash are not fungible for settling an expense shock. In our setting, some obligations require credit, so households keep available credit alongside cash. The motive differs from

precautionary saving, which buffers income risk and depends on prudence (Deaton, 1991; Carroll, 1997): here the shock comes from expenditure, and running out of available credit raises the price of funds.

4.1 Environment

Timing and states The household lives for three dates, $t \in \{0, 1, 2\}$. Dates 0 and 1 fall within one billing month, and date 2 is the next month, when balances settle. Interest accrues once between $t = 1$ and $t = 2$, and the monthly discount factor is β.[15] The household enters date 0 with four state variables:

- $D_0 \geq 0$, the balance carried in from the previous cycle. For a non-revolving household, it sits inside the grace period; for a revolving household, it is interest-bearing.
- $M_0 > 0$, liquid cash, the money available to hand to a courier at the door or money in a checking account.
- $\bar{B} \geq D_0$, the household's credit limit.
- $I \in \{R, N\}$, revolving status. A revolving household carries an interest-bearing balance into the cycle and pays interest rate r on new charges; a non-revolving household pays none. We thus write the effective interest rate as $r_{\text{eff}}(R) = r$ and $r_{\text{eff}}(N) = 0$.

Cash M_0 and the balance D_0 are the outcomes of earlier spending and repayment decisions, and we take them as given.

The payment choice An order of fixed size x arrives, and we condition on it occurring: the household has decided what to buy and chooses only the payment instrument. Writing π for the cash share of the order,

$$M_1 = M_0 - \pi x, \qquad D_1 = D_0 + (1 - \pi)x, \qquad H = \bar{B} - D_1,$$

[15] The repayment is mechanical at the end of $t = 2$. We do not allow immediate repayment at $t = 0$ after paying on credit: in the data, a payment outside the typical repayment window that follows a RappiCard purchase of the same size within three days occurs only 0.2 percent of the time.

where H is the available credit left once the order is paid. The delivery platform accepts one payment method per order, chosen before the order is confirmed, so the choice is binary and $\pi \in \{0, 1\}$; we write $\pi \in [0, 1]$ to keep the derivative in Section 4.2 well defined. For simplicity, we assume that both tenders are feasible for the $t = 0$ purchase: $M_0 > x$ and $\bar{B} - D_0 > x$.

The cost of paying cash Paying cash costs $\eta \geq 0$ per dollar of the order, including the forgone credit rewards, the inconvenience of paying cash for this order, and the value of what else that cash would have paid for. We treat η as a draw from a distribution F at the order level, independent of the household's credit position.[16] Therefore, two orders paid in different ways by two households at the same credit position differ only in their draw of η.

Expenditure shock At $t = 1$, a lumpy expense ξ of random size arrives: an appliance, a car repair, a medical bill, or a trip that has to be booked. Assumption 1 describes its distribution and how it is funded.

Assumption 1 (The credit-financed expenditure shock)**.**

(i) $\xi \sim G$ *with density* g *monotone decreasing on a bounded support* $[0, \bar{\xi}]$*. The baseline is the exponential with rate* λ*, truncated at* $\bar{\xi}$*.*

(ii) A share θ *of the expense falls on credit. For some obligations, the payment technology imposes the restriction: a pre-authorization or deposit at a hotel, a car rental firm, or a private hospital can be placed only on a credit line. For others, arrival circumstances make credit the only feasible way to pay, and the household expects to put the expense on credit. The share* θ *is random across cycles, independent of the size of* ξ*, the credit position, and the household's cash holdings, with* $\mathbb{E}[\theta] > 0$*.*

(iii) A credit-financed amount in excess of available credit H *is funded outside the credit line at the overflow price* $\rho > r_{\mathit{eff}}$ *per dollar, the average cost of whatever covers the*

[16] To explain the observed cash-credit gradient, η would have to fall as the credit line fills. The natural concern runs the other way: a household near its limit is more likely to be short of cash, which raises η and biases the gradient toward zero.

excess, such as credit on a relative's card, informal credit, or paying the expense late at a penalty. Both prices are known at $t = 0$.

Part (i) implies that small shocks are common and large shocks are infrequent. We validate the shape using the 2022 Mexican National Household Expenditure Survey (ENIGH): among households with a credit card, monthly spending in shock-prone categories is close to exponential.[17] With θ independent of size, the credit-financed amount $\theta\xi$ is a rescaling of ξ and inherits its decreasing density.

Part (ii) is a property of the shock, not a preference for credit. The household puts a share θ of the shock on credit because of the payment technology or the circumstances of arrival. Every proposition below holds for each positive value of θ and for the average over the values θ can take. For the derivations, we set $\theta = 1$.

Part (iii) states that an overflow changes the marginal cost of funds without blocking the expense. The household pays the overflow price on the excess and the effective interest rate on the rest.

Settlement At $t = 2$, income y arrives, all balances settle, and the household consumes the residual wealth. When the credit-financed shock can be fully paid with available credit, the household is in the absorbed regime (superscript a):

$$c_2^a = y + M_1 - (1 + r_{\text{eff}})(D_1 + \xi).$$

When the shock exceeds available credit, the card fills to $\bar{B}$, and the excess is funded outside; this is the overflow regime (superscript o):

$$c_2^o = y + M_1 - (1 + r_{\text{eff}})\bar{B} - (1 + \rho)(\xi - H).$$

Utility over consumption is constant relative risk aversion (CRRA) with relative risk aversion γ. Income is high enough that $c_2 > 0$ in both regimes: family support and social transfers provide a floor, so debt service never drives consumption to zero.

[17] The categories are hospital bills, home appliances, car repairs, and funerals, per adult, above 500 pesos (USD 29.5) a month. Expenditures below this threshold are more likely to be routine.

4.2 The total marginal cost of credit (TMCC)

The household chooses the payment tender at $t = 0$ to maximize

$$V(\pi) = -\eta\,\pi x + \beta\,\mathbb{E}[u(c_2)]\,,$$

which is the only choice in the model. Paying the order with cash reduces M_1 and increases H. In the absorbed regime, it saves the interest it would have accrued, and in the overflow regime it saves the overflow price:

$$\frac{\partial c_2^a}{\partial \pi} = r_{\text{eff}}\,x, \qquad \frac{\partial c_2^o}{\partial \pi} = \rho\,x.$$

Differentiating V and splitting the expectation at the regime boundary $\xi = H$,

$$\frac{\partial V}{\partial \pi} = -\eta x + \beta x\Big\{ r_{\text{eff}}\,\mathbb{E}[u'(c_2^a)\mathbf{1}\{\xi \le H\}] + \rho\,\mathbb{E}[u'(c_2^o)\mathbf{1}\{\xi > H\}] \Big\}.$$

Setting $\partial V/\partial \pi = 0$ and dividing by x gives

$$\boxed{\eta \;=\; \underbrace{\beta\, r_{\text{eff}}\,\mathbb{E}[u'(c_2^a)\,\mathbf{1}\{\xi \le H\}]}_{\text{interest}} \;+\; \underbrace{\beta\,\rho\,\mathbb{E}[u'(c_2^o)\,\mathbf{1}\{\xi > H\}]}_{\text{shadow cost}}} \tag{2}$$

The right-hand side is the total marginal cost of credit (TMCC), the expected utility cost of paying a dollar on credit given the credit position. We refer to it as the cost of credit in the text.

The interest term is multiplied by r_{eff}, which is zero for a non-revolving household and r for a revolving one, and by the indicator for the absorbed regime, $\mathbf{1}\{\xi \le H\}$, where the marginal dollar stays on the card and paying cash saves its interest. The shadow cost is multiplied by the overflow price ρ and by the indicator for the overflow regime, $\mathbf{1}\{\xi > H\}$. Charging the order lowers H and makes an overflow more likely, so the shadow cost rises with the balance. Together, a revolving household faces a higher cost of credit than a non-revolving one, and both face a cost that rises with utilization.

Because the choice is binary, (2) is the indifference condition that defines the marginal

order: the household pays cash when η is below the cost of credit.[18] The cash share of orders at a given credit position is therefore $\pi = F(\text{TMCC}(D_0, \bar{B}; I))$, the distribution of the cash cost, evaluated at the cost of credit, so a higher cost of credit raises cash use regardless of the form of F. The curvature of the cash share is not the curvature of TMCC alone: the observed convexity combines the shape of F with the convex cost of credit.

Credit capacity, not wealth The property of (2) that matters for identification is that it prices credit capacity and not wealth. The user fixed effect in Section 3 removes lifetime resources by comparing each household to itself. The remaining concern is transitory: a month of high utilization may also be a month of low income, and a poorer household values a saved dollar more because the utility function is concave. Under risk neutrality, the cost of credit has the closed form

$$\text{TMCC}\big|_{\gamma=0} \;=\; \beta\, r_{\text{eff}} \;+\; \beta\,(\rho - r_{\text{eff}})\, e^{-\lambda H}, \tag{3}$$

which is strictly increasing and strictly convex in D_0 given the decreasing shock density. The gradient therefore does not depend on the curvature of marginal utility, and in this case liquid cash M_0 drops out of the cost of credit altogether: available credit alone generates the shape.

The free-substitution null Suppose the shock at $t = 1$ could be met interchangeably by cash and credit. The total resources available for it are then

$$H + M_1 = M_0 + \bar{B} - D_0 - x.$$

The risk of overflow then depends only on this sum, which does not change with π, the payment choice at $t = 0$. Under this null, only the interest channel matters: cash use by non-revolving households should not respond to utilization until the limit is reached, and a credit

[18] The condition is exact for the binary choice. Consumption is continuous at the regime boundary, so V is differentiable in π and $V(1) - V(0) = \int_0^1 (\partial V/\partial \pi)\, d\pi$. The household pays cash when η is below $\frac{1}{x}\int_0^x \text{TMCC}(D_0 + z)\, dz$, the average cost of credit over the x dollars the order occupies on the line, which the feasibility condition $\bar{B} - D_0 > x$ keeps below the limit. Averaging preserves monotonicity and convexity, so every statement in Section 4.3 holds as written.

limit increase should not change cash use in the absorbed regime. The empirical patterns in Section 3 and the causal results in Section 5 contradict both predictions.[19] Households in non-revolving cycles raise cash use as utilization rises, so they value credit capacity and cash liquidity differently.

4.3 Propositions

The closed-form expression for TMCC yields three propositions about its properties; the full proofs are in Appendix A. The risk-neutral expressions after each proposition use the untruncated exponential distribution for the $t = 1$ shock.

Proposition 1 (Monotonicity)**.** *TMCC is increasing in* D_0*.*

A higher balance lowers available credit dollar for dollar, so the shadow cost rises on both margins: the overflow probability $1 - G(H)$ grows, and consumption in overflow states falls. Under risk neutrality, the interest term is flat, so the rise in TMCC comes from the shadow cost: $\partial \text{TMCC}/\partial D_0 = \beta(\rho - r_{\text{eff}})\lambda e^{-\lambda H} > 0$. Under risk aversion, the interest term also rises slightly, because a higher balance lowers consumption and raises marginal utility.

Proposition 2 (Convexity)**.** *TMCC is convex in utilization* $D_0/\bar{B}$*, given a monotone decreasing shock density and* $\rho > r_{\text{eff}}$*.*

Charging an additional dollar raises the overflow probability by the mass of shocks within that one-dollar interval. At low utilization, this dollar falls in the thin tail of the shock distribution. As a household approaches the limit, the same dollar increases the overflow probability more sharply because the shock distribution is denser near the limit. Under risk neutrality, the second derivative is $-\beta(\rho - r_{\text{eff}})g'(H)$, which is positive whenever g is decreasing. A risk-averse household would have a more convex cost curve.

Proposition 3 (Revolving gap)**.** *At every* D_0*, the TMCC of a revolving household exceeds that of a non-revolving household.*

[19] Appendix A formally states the null hypothesis: under free substitution, the shadow cost vanishes at every level of utilization, and $\partial \text{TMCC}/\partial \bar{B} = 0$ for all degrees of risk aversion. The remaining gradient is purely a wealth effect, which disappears under risk neutrality.

A revolving household pays interest on the marginal dollar, on top of the shadow cost that every household bears, and its consumption in the overflow regime is lower because of the interest debt. The gap in the cost of credit narrows near the limit, where interest becomes a smaller share of the total cost; the gap in cash shares passes through F, which the model does not predict.

4.4 Predictions and tests

The three propositions match the shape of the cash-credit gradient in Section 3, where cash use rises in a convex pattern with credit utilization in non-revolving and revolving months, and revolving households pay cash more often at every utilization. Section 5 uses credit limit increases as a natural experiment to test whether a change in the shadow cost moves payment choice.

5 Credit Limit Increases: Regression Discontinuity Design

5.1 Model predictions

The model predicts that a credit limit increase (CLI) lowers cash use. At a fixed balance D_0, a higher limit $\bar{B}$ raises available credit $H = \bar{B} - D_1$ one-for-one while leaving the balance unchanged, so only the shadow cost falls. Differentiating the cost of credit (3) under risk neutrality with respect to the limit gives

$$\frac{\partial \text{TMCC}}{\partial \bar{B}} \;=\; -\beta\,(\rho - r_{\text{eff}})\,\lambda e^{-\lambda H} \;<\; 0.$$

A credit limit increase does not affect the household's cost of paying cash η or its distribution, so it moves only the right-hand side of the indifference condition $\eta < \text{TMCC}$, and the cash share $\pi = F(\text{TMCC})$ responds through F alone:

$$\frac{\partial \pi}{\partial \bar{B}} \;=\; f(\text{TMCC})\,\frac{\partial \text{TMCC}}{\partial \bar{B}} \;<\; 0,$$

where f is the density of F. The prediction holds over the short period after the limit changes. Over longer horizons, a higher limit feeds back into consumption, repayment, and balances, all of which are taken as given in the model. Therefore, the outcome window for a clean prediction is the first month after the credit limit change.

The convexity of TMCC also predicts that the shadow cost falls more when a household is at higher utilization. Differentiating once more with respect to the balance,

$$\frac{\partial^2 \text{TMCC}}{\partial \bar{B}\, \partial D_0} \;=\; -\beta\,(\rho - r_{\text{eff}})\,\lambda^2 e^{-\lambda H} \;<\; 0.$$

Thus, the model also predicts a more pronounced cash response among households with high utilization, with minimal effect when the credit limit is far from binding.

Furthermore, because the limit increase only moves the shadow cost borne by all households, the model predicts that revolvers should not exhibit a stronger cash response than non-revolvers. We test these predictions using CLIs initiated by RappiCard, where credit score cutoffs are used as an instrument for identification.

5.2 Design

We leverage four CLI campaigns launched between September 2023 and September 2024 that use a credit score of 650 as the cutoff for receiving a limit increase offer. RappiCard initiates all CLIs.[20] Treatment is not determined solely by credit scores: clearing the cutoff significantly increases the chance of an offer but does not guarantee an offer, and a household may decline an offer. Thus, we use a fuzzy regression discontinuity design (RDD) by exploiting variation around the score cutoff. The running variable is a household's credit score at the time of evaluation. The instrument is an indicator that the score exceeds the cutoff. A household is treated if it accepts a credit limit increase.

The outcome is the count of cash-paid Rappi orders in the 30 days after the campaign date. The count sums the per-order choices of the indifference condition over the window. It is defined for every household and equals zero when no order is placed, so the estimate does

[20]The firm notifies households of an offer, and they have 40 days to accept. The firm sometimes ran small pilot campaigns alongside the main campaigns. The firm does not share the full eligibility rules with us.

not condition on post-campaign ordering activity, which is itself an outcome.[21] The median increase is about 3,000 pesos, equivalent to 40 percent of the previous limit. Lastly, because the RDD produces estimates local to the cutoff, the design identifies how households with scores near the cutoff respond.

We use RDHonest inference because the running variable is an integer score that takes few distinct values near the cutoff. RDHonest (Armstrong and Kolesár, 2018, 2020) bounds the second derivative of the conditional mean by a constant M and returns a fixed-length interval valid uniformly over that smoothness class.[22] Throughout the section, the kernel is triangular, the score is centered within each campaign, and the standard errors are clustered at the user level.

Validity The design treats the credit limit as randomly assigned around the score cutoff: households just below the cutoff and just above it should be alike in everything but the chance of receiving a limit increase offer. Two facts support the validity of the design. First, precise sorting onto a given campaign's credit score cutoff is unlikely, since campaign timing and score cutoffs vary across campaigns and are not disclosed. In addition, the density irregularities in credit scores stem from round-number heaping on multiples of five rather than sorting around the cutoff, as Appendix Figure C2 illustrates. Second, to check that predetermined characteristics are continuous at the cutoff, we run a balance check for a range of pre-period variables, including payment choice and spending, repayment and revolving balances, demographics, and credit bureau information. Appendix Table D9 reports the balance check results. All of the 28 predetermined covariates are balanced at the cutoff.

5.3 Results

First stage The first stage examines whether surpassing the cutoff changes the likelihood that a household receives and accepts a CLI. Table 5 presents the first-stage regression

[21] The median number of days between the campaign date and acceptance is 11 days; therefore, to account for the delay in the limit increase taking effect, we use an outcome window in which 99 percent of acceptances are recorded.

[22] Bandwidth selection and bias correction in `rdrobust` (Calonico et al., 2014) are derived for a continuous running variable; with a discrete score the local neighborhood contains few support points, and the asymptotics behind the bias-corrected interval need not apply, so coverage can fall below the nominal rate (Kolesár and Rothe, 2018). Bandwidth selection is automatic in RDHonest.

results, and Figure 6 plots the jump in the share of households accepting an increase against their credit score around the cutoff. Acceptance increases by 18.5 percentage points when the credit score marginally exceeds the cutoff.[23] To run heterogeneity tests for utilization levels and revolving status, we perform separate first-stage regressions for individuals with high (credit utilization at or above 60 percent) versus low utilization prior to the campaigns (Appendix Figure C3), as well as for non-revolvers and revolvers. The jumps are significant in all four subgroups. The differences in jump sizes reflect the issuer extending fewer offers to high-utilization and revolving households, since the take-up rate is around 90 percent across groups. Most increases are fixed amounts: two-thirds of accepters receive 2,000 or 3,000 pesos, and the median increase is 3,000 pesos in both utilization groups. The first-stage estimates remain strong when using the percentage increase in the limit as the outcome variable.

[Figure 6 and Table 5 about here.]

Reduced-form results Figure 7 and Panel A of Table 6 present the reduced-form discontinuity in cash use over the 30 days after the campaign date. Crossing the eligibility threshold reduces cash-paid orders by an average of 0.133. This response is statistically significant among households with utilization rates of 60 percent or higher, where cash-paid orders decrease by 0.140. In contrast, the estimate for households with lower utilization is smaller (-0.058) and lacks precision.

Since cash pays only about one Rappi order in ten, the count is inevitably right-skewed. Figure 7 reports two additional measures of the outcome variable to rule out extreme values driving the results: the inverse hyperbolic sine of the count and an indicator of any cash-paid order. In the high-utilization group, both measures fall significantly at the cutoff: the asinh count by 0.066 and the probability of any cash-paid order in the month by 4.5 percentage points against 10.5 percent just below the cutoff.[24] The cash response of the two alternative outcomes in the high utilization group is significantly larger than that for households with

[23] Per-campaign first-stage F is between 33 and 84, so the instrument is strong and not driven by any single campaign.

[24] We report reduced-form estimates throughout. Dividing by the first stage would imply an effect larger than compliers' own pre-campaign cash use, so we do not interpret the ratio as a per-accepter effect.

low utilization, as confirmed in Appendix Table D10. This pattern aligns with the model prediction that increasing available credit reduces cash use most for households with less available credit. In addition, Appendix Figure C4 reports the jump in Rappi orders for other payment instruments. RappiCard, other-card, and total number of orders do not change significantly, so the reduction in cash usage reflects a shift in the payment mix.

Subsequently, Panel B of Table 6 presents the reduced-form results when splitting households by revolving status before the campaigns. Non-revolvers whose score crosses the cutoff reduce cash-paid orders by 0.128, while revolvers reduce them by 0.144; these estimates are significant at the 10 percent level and we cannot reject equal responses (p-value of 0.87). The results are consistent with the model prediction that households in different revolving statuses should respond similarly.

[Figure 7 and Table 6 about here.]

Robustness The next test assesses whether the cash response occurs at the actual cutoff rather than at arbitrary points within the credit score distribution. We re-estimate the reduced-form regression at false cutoffs offset by $\pm\{10, 15, 20\}$ points, one side of the true cutoff at a time, so the true jump never enters a placebo window. Appendix Figure C5 plots the estimates for all three utilization groups. None of the eighteen false cutoffs yields a negative and significant estimate. The only interval that excludes zero, at -15 points in the high-utilization group, is positive, and its effective sample is less than a tenth of that at the true cutoff.

The estimated effect in the high-utilization group is also robust in alternative specifications, such as the smoothness bound M and the kernel (see Appendix Figure C6 and Appendix Table D10). In addition, holding M fixed, the high-utilization estimate remains negative and significant from half to 1.25 times the optimal bandwidth, equivalent to 8 to 20 points (Appendix Figure C7). The estimate stays negative at wider windows but loses significance because a wider window includes households far from the cutoff, and the honest interval must allow for a larger error from a possibly non-linear relationship between cash use and scores.

Spending response We examine the spending response to limit increases on RappiCard. We compute the increase in credit limit and card spending as a proportion of the previous limit, so regressing the ratio of the two on the cutoff can be interpreted as the marginal propensity to consume from the limit increase. The point estimates are 0.06 for the first month and 0.23 over three months (Appendix Table D11), but the standard errors are too large to infer significance. These estimates align with the range reported in the existing literature (Gross and Souleles, 2002; Agarwal et al., 2018; Aydin, 2022). However, the design lacks sufficient power to detect a significant effect; detecting a propensity of 0.10 needs approximately an order of magnitude more data.

5.4 Paying down the balance

A repayment simultaneously affects both components of the cost of credit: it restores available credit and reduces interest-bearing debt. Therefore, the model predicts that cash use decreases after a repayment, especially when the starting utilization is higher and among revolving households. We test the prediction on Rappi orders placed within 10 days after a voluntary repayment. We regress the indicator for cash payment on utilization reduction by the repayment as a share of the limit, interacted with pre-repayment utilization and revolving status. We control for days since payment and its square, order controls, and fixed effects for user, billing month, and purchase category. Appendix Table D12 presents the estimates. Cash use falls in the ten days after repayment, and falls more when starting utilization is higher. Moreover, a household in revolving status reduces cash use more only when paying down from a higher balance. Because repayment timing and amount are the household's own choices, these estimates are consistent with the model but do not identify a causal effect.

A limit increase moves only the shadow cost, a repayment moves both the shadow cost and the interest cost, and cash use responds to each as the model predicts. The next section turns to estimating the costs of credit and answers what preserving available credit by paying cash costs the household.

6 What a Dollar of Credit Costs

The previous section shows that a credit limit increase reduces cash usage by reducing the shadow cost of credit, as the model predicts. This section prices the interest and shadow cost in cents per dollar of credit spending at each utilization level, separately for revolving and non-revolving households, and values the available credit that households preserve by paying cash.

From observed cash shares at five utilization levels and two revolving states, we recover the cost of credit at each level through the model's indifference condition (2). We measure the interest a dollar on credit accrues until it is repaid or the household stops paying, from the card's repayment records, and the shadow cost is the additional cost the model needs to match the observed cash shares. We treat the observed repayment paths as the household's optimal choices, so changing repayment after one more dollar on credit does not change that dollar's cost to first order. We estimate the model on the RappiCard, since interest and available credit are measured from RappiCard records. The estimated shadow cost is then the probability of overflow times the average cost of covering the expense beyond its available credit on the RappiCard.

6.1 Taking the indifference condition to the data

The indifference condition under risk neutrality We take the risk-neutral form of the indifference condition to the data. Under risk neutrality, marginal utility is constant, and the cost of credit becomes a money price that depends only on the credit position. With a credit-financed share θ of the shock, the indifference condition is

$$\eta \;=\; \underbrace{\beta\, r_{\text{eff}}}_{\text{interest}} \;+\; \underbrace{\beta\,(\rho - r_{\text{eff}})\,\Pr(\theta\xi > H)}_{\text{shadow cost}}. \tag{4}$$

A dollar on credit costs the interest price when the credit-financed amount fits within available credit and the overflow price ρ when it does not, each weighted by its probability.[25]

[25] Risk aversion would not change the total cost of credit, which the cash shares determine. It would move part of the cost from the shadow cost to interest and lower the overflow price needed to explain the same cash shares.

Credit-financed shock A shock falls on credit in one of two ways. Both draw their size ξ from the truncated exponential G of Assumption 1(i), with mean 3,000 pesos (about USD 177) and cap $\bar{\xi} = 30{,}000$ pesos, calibrated to the Mexican Household Expenditure Survey (ENIGH) 2022 shock-prone expenditure per adult per month.[26] The first we observe in the RappiCard records. With probability p, the household pays a hotel, car rental firm, private hospital or travel agency, which places a pre-authorization hold on credit for a share θ_m of the expense. The second is every other expense that requires credit, which card records cannot separate from purchases charged by preference. With probability q, the whole expense requires credit. The expected share of a shock that requires credit is therefore

$$\mathbb{E}[\theta] \;=\; p\,\theta_m \;+\; q. \tag{5}$$

We measure p and θ_m from RappiCard transactions at these merchants (Appendix B.1), and estimate q as the value at which the model's cash shares match the observed ones.

The cost of credit Substituting the two shocks into (4) gives the cost of credit for a household with revolving status I at utilization u:

$$\mathrm{TMCC}^I(u) \;=\; \underbrace{\kappa^I(u)}_{\text{interest}} \;+\; \underbrace{\big(\rho - \kappa^I(u)\big)\Big[\,p\,\Pr\big(\theta_m\,\xi > H\big) + q\,\Pr\big(\xi > H\big)\Big]}_{\text{shadow cost}}, \qquad H = \bar{B}\,(1-u), \tag{6}$$

where $\kappa^I(u)$ is the measured interest price, which replaces r_{eff}, and ρ is the overflow price of Assumption 1(iii). The bracket replaces $\Pr(\theta\xi > H)$ in (4): it is the probability that a shock paid on credit exceeds H. Both prices are cumulative costs per dollar over the life of the debt. We set $\beta = 1$: both prices are undiscounted lifetime costs, so a common discount factor rescales them alike and leaves the comparison unchanged.

We measure the interest component from the revolving-balance ledger of RappiCard. The interest price counts the interest a household pays on a dollar of debt: we track each dollar

[26] Shock-prone expenditure covers health, appliance and home repair, vehicle repair and funeral spending. We fit the rate on households spending more than 500 pesos per adult in a month, where such spending is less likely to be routine. The fitted mean is 3,149 pesos, which we round to 3,000, and a gamma fit with a free shape parameter yields a shape close to an exponential distribution. The cap at ten times the mean removes less than 1 percent of the distribution.

originated in the first 12 months after card approval until it is repaid or the household stops paying, and record the interest it accrues over that duration. Averaging this interest per dollar over the revolving cycles in each of five utilization bins gives $\kappa^R(u)$, constant within a bin. A household in a non-revolving cycle pays no interest on the purchase today, but it may become a revolver next month, with a probability that depends on its utilization. Its interest price $\kappa^N(u)$ is $\kappa^R(u)$ times that probability, measured from the data.[27] The shadow cost is not measured and varies continuously with utilization, so we evaluate it at each order's available credit on the day before the order.

The observation equation The indifference condition imposes that an order is paid in cash when its cash cost η falls below the cost of credit. Because some orders are paid in cash even in cycles when little credit is used, we decompose the cash share into a share π_0 of orders that go to cash at any price, and the share of the remaining orders whose cash cost lies below TMCC:

$$\pi^I(u) \;=\; \pi_0 + (1-\pi_0)\, F\big(\mathrm{TMCC}^I(u)\big). \tag{7}$$

We estimate π_0 and F jointly with q and ρ, matching (7) to the fitted cash shares in five utilization bins for each revolving status from column 3 of Table 2. The estimation is therefore overidentified, with 10 data moments and 5 free parameters to estimate.

Assumptions The mapping from cash shares to the cost of credit extends the assumptions from the conceptual framework. First, the cash cost η is independent of the credit position, so the cash share moves with utilization only through TMCC. Second, credit-financed shocks are exogenous to the credit position: p, q, θ_m and G do not vary with utilization or revolving status, so the overflow probability changes with utilization only through H. Lastly, households in revolving and non-revolving cycles share the same π_0 and F, so the gap between their cash shares at a given utilization reflects the gap in their cost of credit. The cash shares we match come from comparisons within households, so persistent differences in cash cost across households do not enter them, and the assumption requires only that a household's

[27] Appendix B.1 explains how we follow a dollar of debt and measure the probability that a non-revolving cycle turns revolving.

cash cost does not change with its revolving status. If a household starts to revolve in months when cash is scarce, its cash cost is higher in revolving cycles, the observed gap understates the response to the interest gap, and the shadow cost is overstated.

6.2 Measurement, estimation and fit

Measured interest Table 7 presents the measured inputs and estimated parameter values in Panels A and B. The interest price of a dollar of revolving debt, $\kappa^R(u)$, is hump-shaped in utilization. We find the hump shape comes from the duration of revolving debt rather than the accrued rate it faces: a dollar originated in the 20 percent to 40 percent utilization bin stays the longest at 5.6 months, and balances near the limit are more often written off. The interest price of a non-revolving dollar, $\kappa^N(u)$, rises with utilization instead, because the probability that a non-revolving cycle turns revolving at the next statement rises from less than 10 percent to 30 percent as utilization increases. Thus, the interest gap between revolving and non-revolving households narrows near the limit.

Estimation We fit the five parameters jointly by classical minimum distance to the ten cash shares, weighting each moment by the inverse of its variance, and bootstrap the intervals. Because the two revolving types share π_0 and F, the gaps between their cash shares in the five utilization bins show how cash use responds to the measured interest gap, which fixes both the monetary scale and the curvature of F. Given F, the rise in cash shares across utilization levels identifies the total cost of credit, and the shadow cost is what remains after the measured interest cost.

Panel B of Table 7 reports the estimates. A household with no available credit left expects to pay 43 cents at the overflow price per dollar of expenditure shock ($\mathbb{E}[\theta]\rho = 0.434$, 95 percent interval [0.312, 0.510]). The expected credit-financed share behind this product, $\mathbb{E}[\theta]$, is 0.13, with a 95 percent interval [0.05, 0.32].[28] ENIGH 2022 bounds this share from above: card-holding households put 37 percent of their shock-prone spending on a credit card. The estimated F is a Weibull distribution that is convex over the range of costs of credit.

[28] Appendix B.3 explains why the interval is wide.

[Table 7 about here.]

Fit The model fits eight of the ten cash shares closely. Figure 8 plots the model's cash shares against the fitted cash shares from the regression. The two exceptions are revolving cycles in the two lowest utilization bins. The measured interest price of a revolving dollar rises by six cents between these bins, yet a household in a revolving cycle at 20 to 40 percent utilization pays cash no more often than when its utilization is below 20 percent. These two cells account for most of the misfit, and the overidentification test rejects at the 5 percent level ($J = 9.80$ with four degrees of freedom, $p = 0.04$).[29] They barely move the shadow cost. Following Andrews et al. (2017), matching both cells lowers the objective sharply, but moves $\mathbb{E}[\theta]\rho$ by only 0.01, from 0.434 to 0.447, well within its interval. When we measure the interest price over the entire panel instead of the first 12 months, it is flatter across these bins, and the overidentification test does not reject in this specification ($J = 5.89$, $p = 0.21$).

[Figure 8 about here.]

Robustness The product $\mathbb{E}[\theta]\rho$ stays inside its 95 percent interval when F is lognormal or log-logistic, when shock sizes scale with the household's aggregate card limit, and when we fit households whose RappiCard accounts for most of their card limits separately from the rest. Appendix B.5 reports each check, and Appendix B.4 reports the recovery of the estimates from simulated data.

6.3 The cost of credit by utilization

TMCC schedule Figure 9 plots the estimated TMCC against credit utilization for revolving and non-revolving cycles. The cost of credit rises with utilization, and more steeply as available credit runs out. The cumulative cost of a dollar on credit is about 8 cents at 20 percent utilization and 31 cents at 80 percent for a non-revolving household, compared to about 32 and 48 cents, respectively, for a revolving household. The gap between the two

[29] The test refits the model with the full covariance of the cash shares as the weight matrix (Appendix B.2). The fixed effects in the cash-share regression absorb one direction of that covariance, which has rank nine, so the test has four degrees of freedom.

types narrows near the limit, because the measured interest gap closes there and a smaller share of each dollar stays on credit.

[Figure 9 about here.]

Decomposition of TMCC Figure 10 decomposes the cost of credit into interest and shadow cost, for non-revolving households in Panel A and revolving households in Panel B. In revolving cycles, interest accounts for the majority of the cost of credit over most of the utilization range. In non-revolving cycles, the shadow cost is 5 of the 8 cents of total cost at 20 percent utilization and 24 cents of the 31 cents at 80 percent.

[Figure 10 about here.]

6.4 What it costs households

Table 8 applies the estimated cost of credit at the household's credit position to the order's value, and reports the cost of credit per dollar they would have carried if these orders were charged. Over the 12 months after card approval, 25,366 households in the estimation sample paid 23.64 million pesos (about USD 1.4 million) in cash on Rappi that could have been charged on credit. For an average non-revolving household, the cost of credit is 21.0 cents per dollar, given the utilization level when it pays cash, of which 78 percent is the shadow cost. If all cash-paid orders were paid on credit, their annual cost of credit would equal a third of their cash spending on Rappi. The estimated cash cost is 9 cents per dollar, including the 3 percent cashback forgone among the always-cash orders. Therefore, the cost of credit exceeds the cash cost by at most 26 cents per dollar on average.

[Table 8 about here.]

6.5 Counterfactuals: credit limit and interest rate changes

The credit limit and the interest rate are the two contract features that issuers and regulators act on. Using the model estimates, we can assess their potential impacts on households' cost of credit and payment choice, conditioning on revolving status and credit utilization. We

compute the change in the cost of credit and in cash use under three changes in the contract: a 10 percent increase in the credit limit, a 10 percent cut in the credit limit, and a 10 percent cut in the contracted interest rate.

The unit of the experiment is a user-month with positive purchases on RappiCard, priced at the household's utilization at that month's statement and its revolving status in that month. The change in the cost of credit in a utilization bin is therefore what a household gains or loses if the change arrives in a month it spends in that bin, and a household enters a bin once for each such month. We hold the household's repayment and consumption fixed. Table 9 reports the change in the cost of credit, in cents per dollar charged to the card, by utilization bin and revolving status.

[Table 9 about here.]

Limit increase A limit change shifts the overflow probability and moves only the shadow cost. A 10 percent limit increase lowers the cost of credit by about 11 cents per dollar in the top utilization bin and by less than 1 cent below 20 percent utilization: the cost curve is convex, so a dollar of added available credit lowers the overflow probability most near the limit. The reduction is of similar size for revolving and non-revolving households. Through the estimated F, a 10 percent limit increase lowers the share of Rappi orders paid in cash by 0.6 percentage points on average, from a baseline cash share of 8.8 percent. The literature reads the rise in card balances after a limit increase as more borrowing and more consumption (Gross and Souleles, 2002; Agarwal et al., 2018; Aydin, 2022). Our estimates suggest that part of this rise comes from payment choice: a larger limit makes the card cheaper to use, so households move purchases from cash to credit without borrowing or consuming more.

Limit cut A limit cut raises the cost of credit more than an increase of the same size lowers it, and the difference grows as utilization approaches the limit. For revolving households, the rise in the cost of credit from a 10 percent cut is 25 percent larger than the fall from a 10 percent increase below 20 percent utilization, and 50 percent larger at 60 to 80 percent utilization.[30] After the 2008 crisis, banks cut credit card limits by more than a quarter

[30] In the top utilization bin, a 10 percent cut raises the cost of credit by less than a 10 percent increase lowers it. The cut leaves many households in this bin with no available credit, and the shadow cost cannot

(Fulford, 2015), and the cuts were larger and more persistent for consumers who could not replace the lost credit with other cards (Chava et al., 2023). Our estimates imply these cuts hit households already near their limit hardest.

Interest rate cut A rate cut lowers the interest price and leaves the overflow probability unchanged. A 10 percent cut in the contracted rate, 0.62 percentage points per month at the median rate, lowers a revolving household's cost of credit by 2 to 3 cents per dollar at every utilization level. The gain scales with how long a dollar stays on credit, so it is nearly zero for a non-revolving household. If a lower rate lengthens spells of revolving debt, as Gross and Souleles (2002) suggest, the gain is likely smaller.

In conclusion, interest rate and credit limit changes reach distinct households through two cost channels, so these two policies do not have comparable outcomes. A rate cut reaches only households that pay interest; a limit change reaches every household, and its impact depends on how close households are to the limit. In addition, the counterfactual shows a response on the payment margin that balance data alone cannot see, implying that some changes in card balances result from households shifting payment methods.

7 Conclusion

This paper measures the price households pay to preserve available credit by using cash in their payment choices. Linking 1.5 million Rappi orders to the household's credit position on the day before each order, we find that the same household pays cash more often as its utilization rises, and the pattern is present even when the household pays no interest. A three-period model explains the pattern with one mechanism: some future expenses can be paid only with credit. Each dollar charged reduces available credit and increases the chance of paying a higher cost to cover such expenses. Hence, a dollar on credit carries a shadow cost beyond interest that grows with utilization. The model predicts that a credit limit increase lowers the shadow cost and reduces cash usage, and we confirm it with a regression discontinuity design: higher credit limits reduce cash payments, most among households with

rise above its level at zero available credit. Instead, the cut takes from these households purchases that no longer fit on the card, a loss of consumption the model does not price.

high utilization, and similarly among revolving and non-revolving households. The model estimates a shadow cost of 5 cents per dollar at 20 percent utilization and 24 cents at 80 percent for a household that pays no interest, close to the 25 cents of cumulative interest on a dollar of revolving debt. The estimate explains the tendency to use cash even when credit appears to be free: a household that never pays interest still faces a cost of credit comparable to a revolving household's when it nears the credit limit.

Our estimates take the household's repayment and consumption as given, while both margins likely respond to the same shadow cost. A household near its limit may repay more to restore available credit or spend less, and the duration of revolving debt is itself a repayment choice. A dynamic model with endogenous repayment and consumption would let the shadow cost act on all three margins at once. It would also connect the price we measure to the value of liquidity of households holding cash alongside credit card debt.

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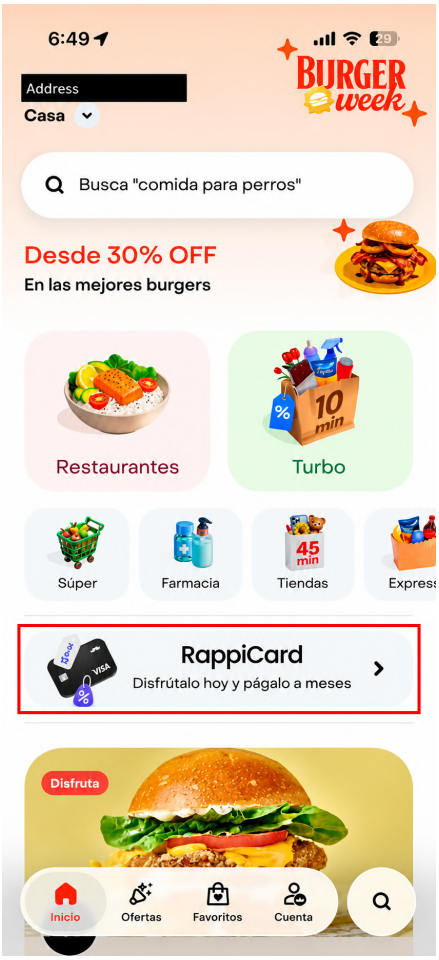
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Panel A: Rappi home screen



Panel B: Payment method at checkout

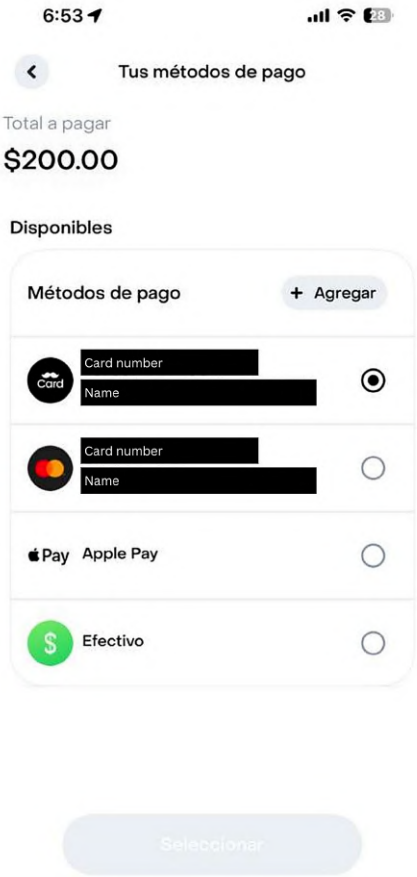


Figure 1: RappiCard and the payment choice in the Rappi app

Notes. This figure shows screenshots from the Rappi app. Panel A is the home screen, with the entrance to the RappiCard page boxed. Panel B is the list of payment methods at checkout: linked cards, Apple Pay, and cash on delivery (*Efectivo*). Card numbers and names are redacted.

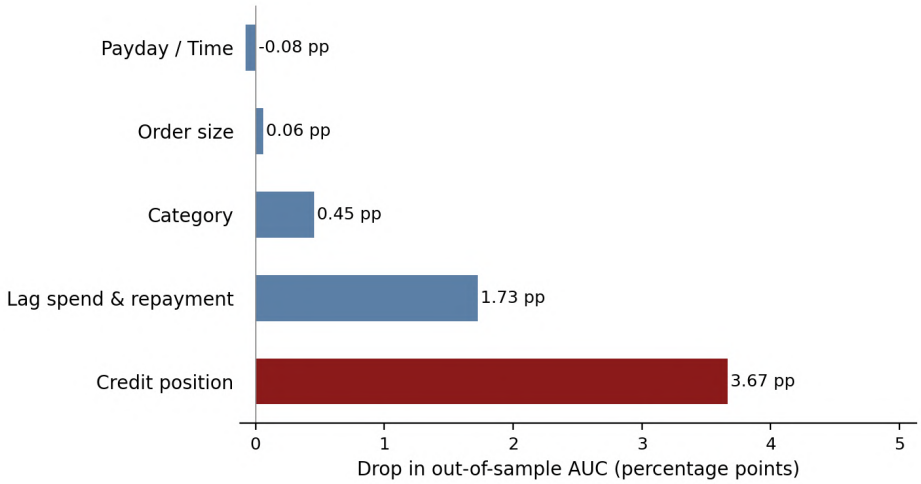


Figure 2: Leave-one-block-out importance for predicting cash payment

Notes. This figure plots the drop in out-of-sample area under the curve (AUC), in percentage points, when one block of covariates is removed from a logistic regression of the cash indicator on the five blocks described in Section 2: credit position, purchase category, timing, prior-cycle spending and repayment, and order size. Every model also includes calendar-month dummies and the household's mean cash rate in the training folds. AUC is averaged over five folds that split households into groups, fitting on four and predicting the fifth; the full model's AUC is 0.612. The sample is orders in unblocked months with day-before utilization at or below 105 percent.

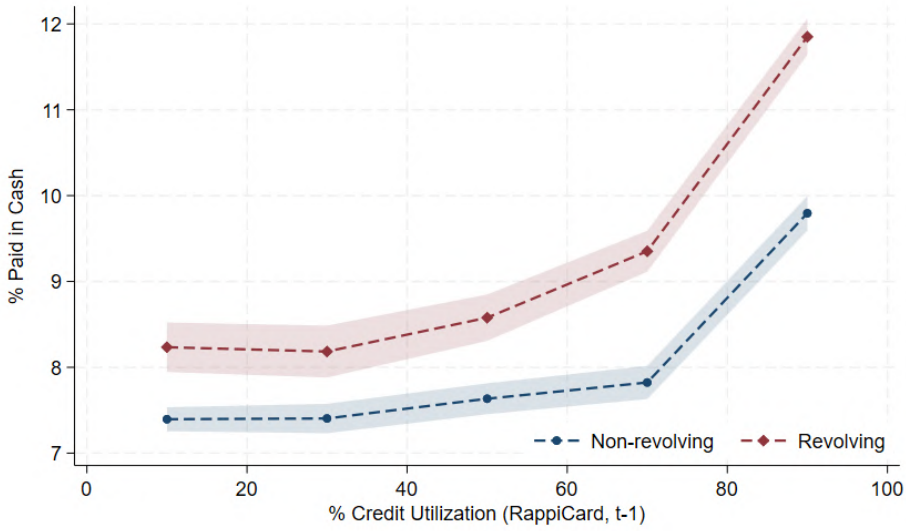


Figure 3: Fitted cash share from within-person cash-credit gradient

Notes. This figure plots the fitted share of orders paid in cash, in percent, at the midpoints of the five utilization bins, from column 3 of Table 2, separately for revolving and non-revolving months. Each point is the average of the regression's fitted values over the orders in the bin and revolving state. Shaded bands are 95 percent confidence intervals from standard errors clustered by user.

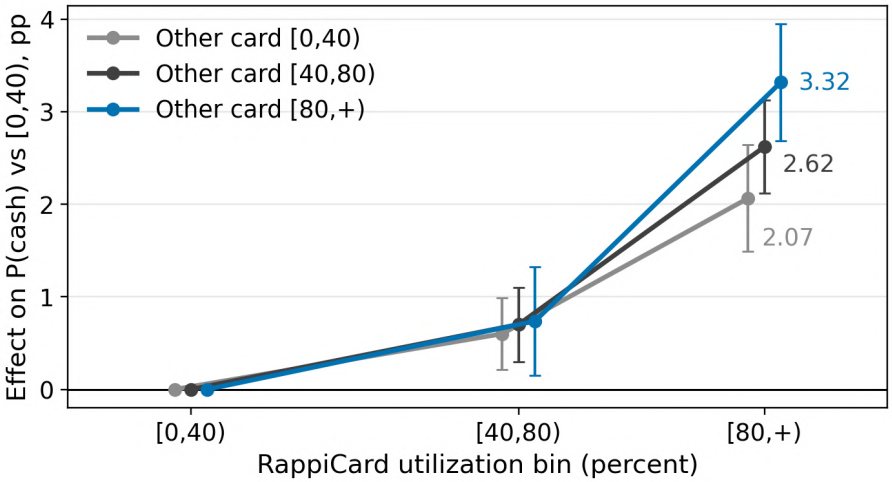


Figure 4: The cash-credit gradient by other-card utilization

Notes. This figure plots the effect of RappiCard day-before utilization on the probability of paying cash, in percentage points, from column 3 of Table D3, at each level of other-card utilization. Both utilizations are in three bins, $[0, 40)$, $[40, 80)$, and $[80, +)$, with $[0, 40)$ omitted. Each point is the own-card bin coefficient plus its interaction with the other-card bin, so each line is the own-card gradient relative to $[0, 40)$ holding the other-card bin fixed, and the $[0, 40)$ points are zero by construction. The regression includes user, bill-month, category, and cycle-position fixed effects, the controls of Table 2, and the log of the household's credit limits on other cards. The sample is orders with a credit bureau pull at most one month old, a positive limit on other cards, and other-card utilization at or below 110 percent. Bars are 95 percent confidence intervals from standard errors clustered by user.

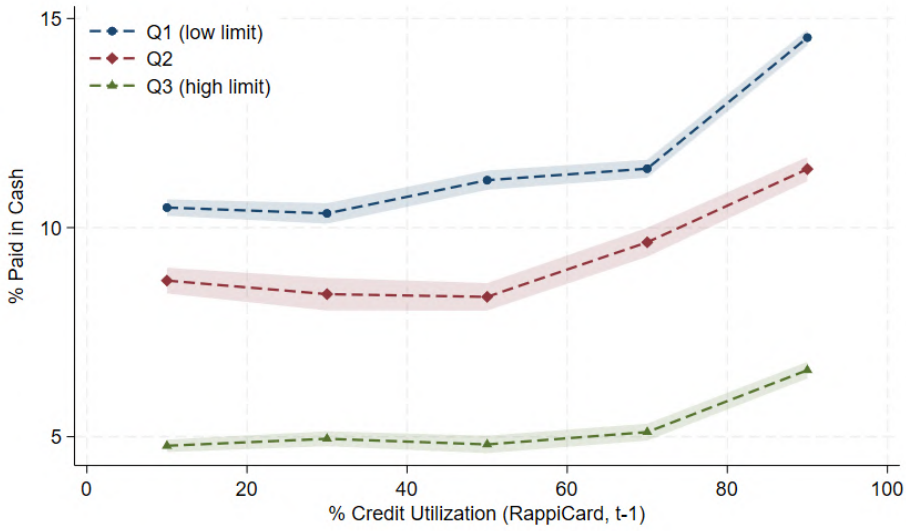


Figure 5: The cash-credit gradient by credit limit terciles

Notes. This figure plots the fitted share of orders paid in cash, in percent, at the midpoints of the five utilization bins, estimated separately within terciles of the household's initial RappiCard credit limit. Within each tercile, the regression is equation (1) without the revolving terms, with the controls and fixed effects of column 3 of Table 2, so each point pools revolving and non-revolving months and is the average of the fitted values over the orders in the bin. The sample is the regression sample of Section 2, excluding orders with day-before utilization above 105 percent. Shaded bands are 95 percent confidence intervals from standard errors clustered by user.

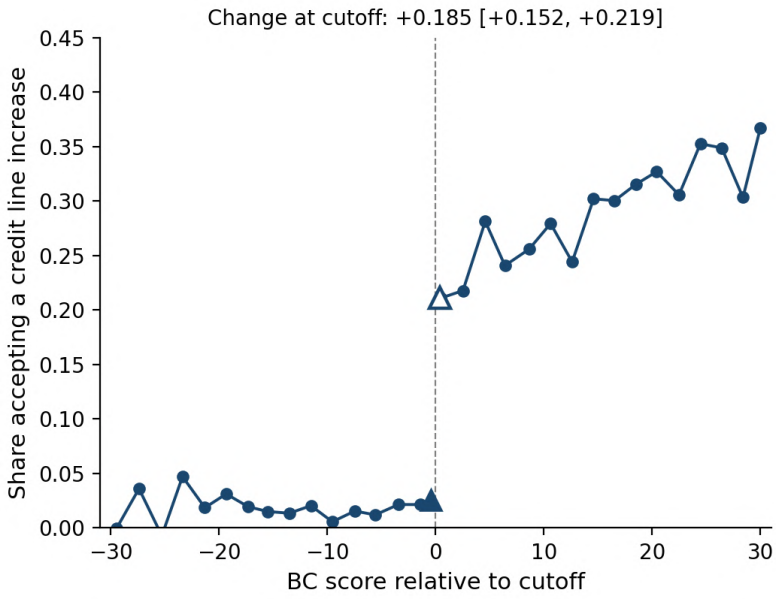


Figure 6: First stage: probability of accepting a credit limit increase

Notes. This figure plots the share of households that accept a credit limit increase against the credit score, centered at each campaign's cutoff, on the sample of Table 5. Acceptance is adjusted for campaign fixed effects. Circles are means in two-point bins within 30 points of the cutoff. Triangles are the local linear predictions at the cutoff from below (solid) and from above (hollow), with a triangular kernel and a bandwidth of 12.1 points. Their difference is the pooled estimate in Table 5, shown above the panel with its honest 95 percent confidence interval.

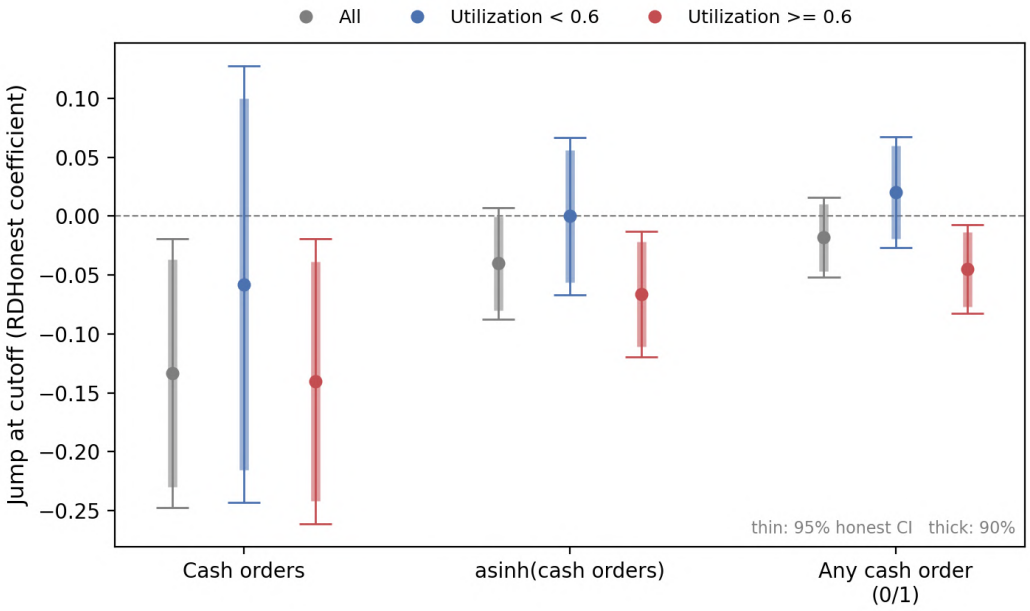


Figure 7: Reduced form: cash response to limit increase, by pre-campaign utilization

Notes. This figure plots the change at the credit score cutoff in three measures of cash use in the 30 days after the campaign: the count of cash-paid Rappi orders, its inverse hyperbolic sine, and an indicator for any cash-paid order, each in its own units. Each measure is estimated for all households and separately for households below and at or above 60 percent utilization before the campaign, by RDHonest with campaign fixed effects and a triangular kernel, on the sample of Table 5. Thin capped lines are honest 95 percent confidence intervals and thick bars honest 90 percent intervals, from standard errors clustered by user.

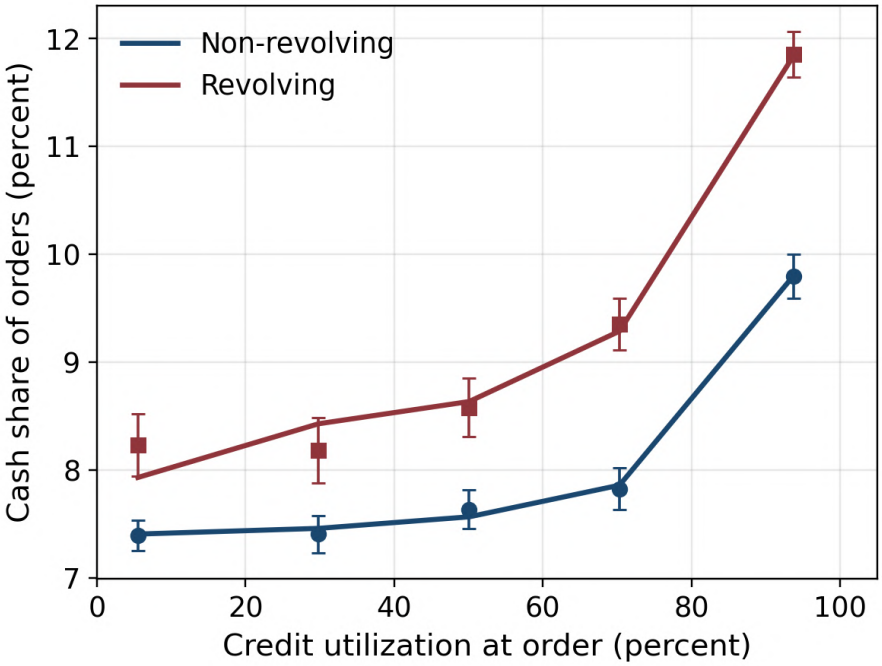


Figure 8: Cash share by utilization: data and model

Notes. This figure plots the share of orders paid in cash by utilization bin in the data and in the model, separately for revolving and non-revolving cycles. Markers are the fitted cash shares from column 3 of Table 2, with 95 percent confidence intervals. Lines are the model's cash shares at the estimates in Table 7, computed order by order at the order's utilization and limit on the day before the order and averaged within the bin.

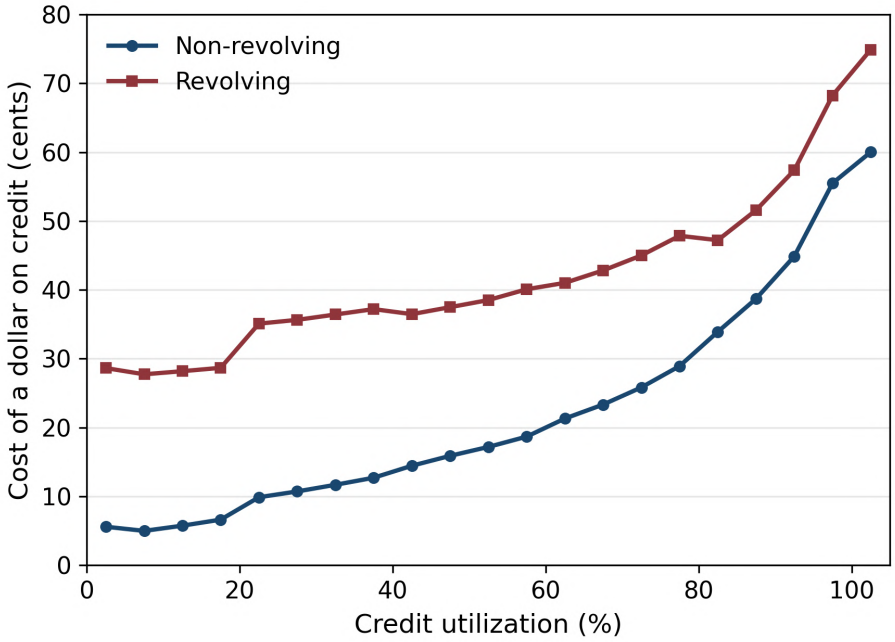
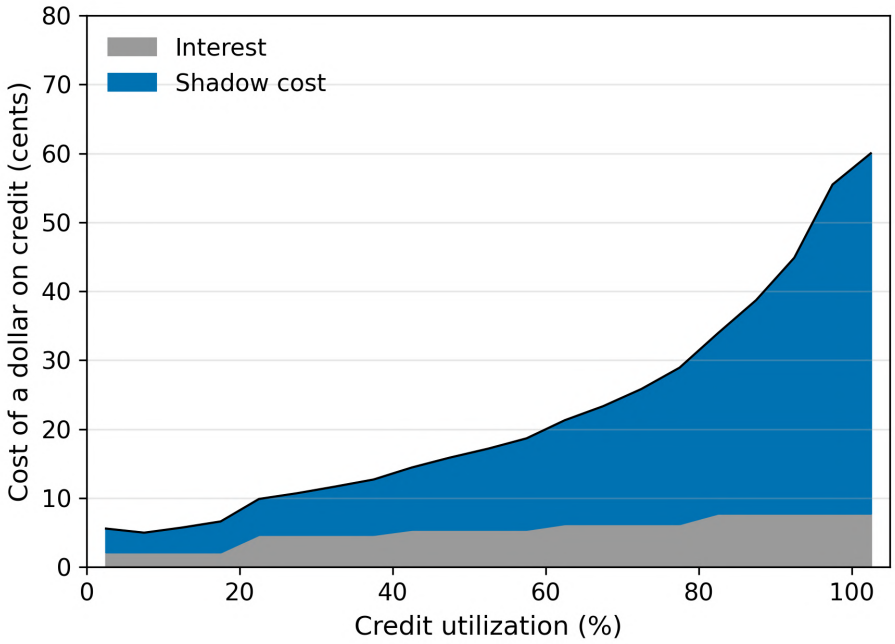


Figure 9: Cost of a dollar on credit by utilization

Notes. This figure plots the cost of credit $\text{TMCC}^I(u)$ of equation (6), in cents per dollar of card spending, against utilization for revolving and non-revolving cycles, at the point estimates in Table 7. The shadow cost is computed order by order at the order's available credit on the day before the order and averaged within 5-point utilization cells. The interest price $\kappa^I(u)$ is measured in five 20-point utilization bins, producing steps in both schedules at the bin edges.

Panel A. Non-revolving cycles



Panel B. Revolving cycles

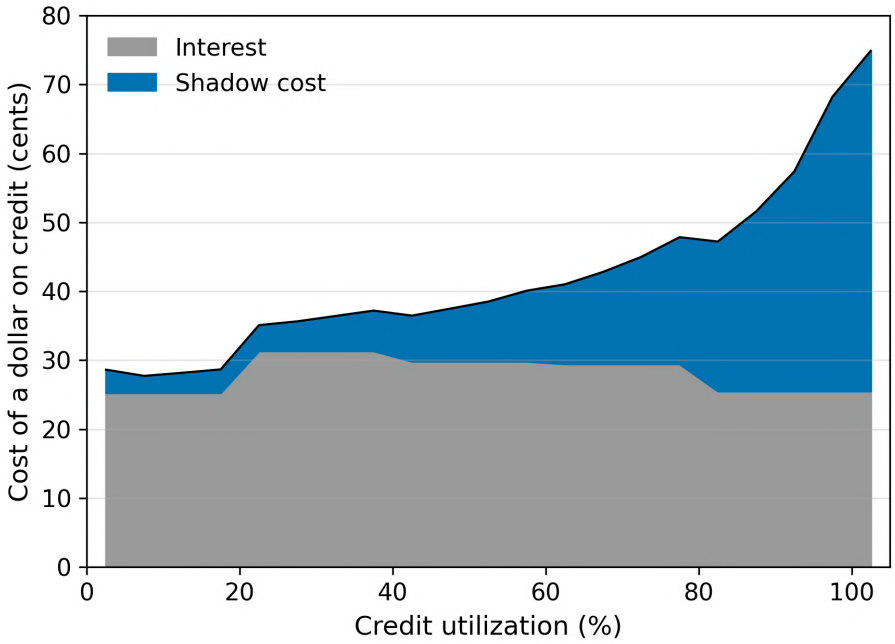


Figure 10: Interest and shadow cost

Notes. This figure plots the cost of credit of Figure 9 split into the measured interest price $\kappa^I(u)$ and the shadow cost, for non-revolving cycles in Panel A and revolving cycles in Panel B, in cents per dollar of card spending, at the point estimates in Table 7. The shadow cost is computed order by order at the order's available credit on the day before the order and averaged within 5-point utilization cells.

Table 1: Summary statistics

	N	Mean	Median	p25	p75
Panel A: User level (72,632 users)					
Female (%)	61,986	39.3	0.0	0.0	100.0
Age	65,329	30.9	28.5	23.8	35.6
Credit history (months)	67,887	107.8	90.4	36.4	164.7
Bureau score (BC)	67,924	655	655	638	672
Has another credit card, at approval (%)	58,276	90.9	100.0	100.0	100.0
Credit limit on other cards, at approval (MXN)	58,292	74,805.9	20,600.0	5,400.0	65,001.0
Total credit card debt, at approval (MXN)	58,292	21,027.9	4,812.0	329.0	18,964.5
RappiCard credit limit, initial (MXN)	72,590	10,827	5,000	5,000	12,000
RappiCard credit limit, last observed (MXN)	72,590	12,132	7,000	5,000	13,000
Number of Rappi orders (12 mo.)	72,632	20	8	3	21
Total Rappi spend (12 mo., MXN)	72,632	8,057.6	1,701.3	555.2	5,154.0
Ever paid cash (%)	72,632	42.6	0.0	0.0	100.0
Ever paid cash while revolving (%)	72,632	25.3	0.0	0.0	100.0
Panel B: User-month level (745,297 user-months)					
In revolving status (%)	745,297	44.5	0.0	0.0	100.0
Interest-bearing balance, if revolving (MXN)	324,764	6,912.8	4,782.9	2,447.8	8,142.5
Interest and fees charged (MXN)	702,243	178.5	0.0	0.0	241.2
RappiCard credit utilization (%)	745,267	56.3	62.0	17.8	94.5
Aggregate credit utilization (%)	340,141	58.2	58.5	31.1	85.4
Panel C: Order level (1,479,751 orders)					
Order value (MXN)	1,479,751	395.5	207.8	145.8	323.1
Number of items	1,479,751	5	2	1	5
Paid by RappiCard (%)	1,479,751	59.2	100.0	0.0	100.0
Paid by other credit card (%)	1,479,751	28.3	0.0	0.0	100.0
Paid by cash (%)	1,479,751	9.5	0.0	0.0	0.0
Paid by other method (%)	1,479,751	3.1	0.0	0.0	0.0
Cash orders placed while revolving (%)	139,992	50.4	100.0	0.0	100.0

Notes. This table reports summary statistics for the estimation sample of Section 2: 72,632 cardholders with at least one Rappi order matched to a statement in the 12 months after card approval. Panel A reports user-level characteristics. Rows marked "at approval" come from the application file, and the two RappiCard limit rows are the first and last limits observed in the window. Panel B reports statistics at the user by billing-month level and Panel C at the order level. Revolving status equals one in billing months in which the account carries an interest-bearing balance. RappiCard utilization is the statement balance over the limit; aggregate utilization adds the other cards at the most recent credit bureau pull and is defined for months with a pull at most one month old. Indicators are in percent and amounts in Mexican pesos. N varies across rows because of missing values.

Table 2: Cash use and RappiCard utilization

	Indicator for cash payment			
	(1)	(2)	(3)	(4)
Credit utilization, % (omitted bin [0,20))				
[20,40)	0.0005	0.0008	0.0006	0.0018
	(0.0010)	(0.0010)	(0.0010)	(0.0013)
[40,60)	0.0029**	0.0035***	0.0030***	0.0052***
	(0.0011)	(0.0011)	(0.0011)	(0.0015)
[60,80)	0.0050***	0.0055***	0.0050***	0.0071***
	(0.0012)	(0.0012)	(0.0012)	(0.0016)
[80,+)	0.0250***	0.0256***	0.0250***	0.0265***
	(0.0013)	(0.0013)	(0.0013)	(0.0017)
Revolving	0.0077***	0.0078***	0.0080***	0.0086***
	(0.0017)	(0.0017)	(0.0017)	(0.0021)
[20,40) × Revolving	-0.0009	-0.0006	-0.0005	-0.0024
	(0.0021)	(0.0021)	(0.0021)	(0.0026)
[40,60) × Revolving	0.0015	0.0008	0.0010	-0.0015
	(0.0022)	(0.0022)	(0.0022)	(0.0027)
[60,80) × Revolving	0.0073***	0.0065***	0.0066***	0.0029
	(0.0022)	(0.0022)	(0.0022)	(0.0028)
[80,+) × Revolving	0.0120***	0.0110***	0.0109***	0.0053**
	(0.0022)	(0.0022)	(0.0022)	(0.0027)
Controls	No	Yes	Yes	Yes
Bureau controls	No	No	No	Yes
Category FE	No	No	Yes	Yes
Cycle-position FE	No	No	Yes	Yes
N	1,279,883	1,260,519	1,255,790	810,411
R^2	0.398	0.398	0.398	0.416

Notes. This table reports estimates of equation (1). The dependent variable is an indicator that the order was paid in cash. Bins are defined on RappiCard utilization on the day before the order, with $[0, 20)$ omitted. Revolving equals one if the account carried an interest-bearing balance at the most recent statement before the order. The sample is the regression sample of Section 2, excluding orders with zero value or with day-before utilization above 105 percent. Column 4 keeps orders with credit bureau information. All columns include user and bill-month fixed effects. Controls are the log of order value, the log of RappiCard purchases in the previous billing cycle, and an indicator for no purchases in that cycle. Bureau controls include the log of the household's credit limits on other tradelines, an indicator for no other limit, the number of active cards, and indicators for a mortgage, a bank tradeline, a microfinance tradeline, and any other delinquent or charged-off tradeline. Cycle-position fixed effects are the five-way split of the billing cycle defined in Section 3. Standard errors clustered by user in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Cash use and aggregate borrowing capacity

	Indicator for cash payment			
	(1)	(2)	(3)	(4)
Aggregate credit utilization, % (omitted bin [0,20))				
[20,40)	-0.0009	-0.0010	-0.0012	-0.0025
	(0.0020)	(0.0021)	(0.0021)	(0.0022)
[40,60)	0.0022	0.0023	0.0021	0.0002
	(0.0024)	(0.0024)	(0.0024)	(0.0026)
[60,80)	0.0065**	0.0064**	0.0063**	0.0058**
	(0.0027)	(0.0027)	(0.0027)	(0.0028)
[80,+)	0.0179***	0.0180***	0.0179***	0.0178***
	(0.0029)	(0.0029)	(0.0029)	(0.0031)
Revolving	0.0078**	0.0077**	0.0076**	0.0082**
	(0.0038)	(0.0039)	(0.0039)	(0.0039)
[20,40) × Revolving	-0.0006	-0.0005	-0.0002	-0.0008
	(0.0044)	(0.0044)	(0.0044)	(0.0046)
[40,60) × Revolving	0.0024	0.0021	0.0023	0.0021
	(0.0044)	(0.0045)	(0.0045)	(0.0046)
[60,80) × Revolving	0.0040	0.0042	0.0044	0.0026
	(0.0045)	(0.0045)	(0.0045)	(0.0046)
[80,+) × Revolving	0.0140***	0.0133***	0.0133***	0.0096**
	(0.0047)	(0.0047)	(0.0047)	(0.0048)
Controls	No	Yes	Yes	Yes
Bureau controls	No	No	No	Yes
Category FE	No	No	Yes	Yes
Cycle-position FE	No	No	Yes	Yes
N	643,974	637,097	634,676	593,803
R^2	0.429	0.428	0.429	0.425

Notes. This table reports estimates of equation (1) with bins defined on aggregate utilization across all of the household's credit cards, as defined in Section 2, with $[0, 20)$ omitted. Revolving equals one if the account carried an interest-bearing balance at the most recent statement before the order. The sample includes orders from the regression sample for which aggregate utilization is defined, which requires a credit bureau pull no older than one month or no other card, and excludes orders with zero value or aggregate utilization above 105 percent. All columns include user and bill-month fixed effects. Controls are the log of order value, the log of RappiCard purchases in the previous billing cycle, an indicator for no purchases in that cycle, and an indicator that another card is delinquent. Bureau controls include the number of active cards and indicators for a mortgage, a bank tradeline, a microfinance tradeline, and any other delinquent or charged-off tradeline. Cycle-position fixed effects are the five-way split of the billing cycle defined in Section 3. Standard errors clustered by user in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Cash-credit gradient and payday window

	Indicator for cash payment		
	(1) Payday	(2) +CU bins	(3) x CU bins
Payday window	-0.0011*	-0.0010*	-0.0006
	(0.0006)	(0.0006)	(0.0009)
Utilization [20,40)		0.0007	0.0009
		(0.0009)	(0.0009)
Utilization [40,60)		0.0039***	0.0039***
		(0.0010)	(0.0011)
Utilization [60,80)		0.0084***	0.0085***
		(0.0011)	(0.0011)
Utilization [80,+)		0.0317***	0.0319***
		(0.0011)	(0.0012)
Utilization [20,40) × Payday window			-0.0012
			(0.0016)
Utilization [40,60) × Payday window			-0.0004
			(0.0017)
Utilization [60,80) × Payday window			-0.0001
			(0.0018)
Utilization [80,+) × Payday window			-0.0007
			(0.0015)
p: all bins × payday			0.9593
p: top bin × payday			0.6519
N	1,260,519	1,260,519	1,260,519
R^2	0.397	0.398	0.398

Notes. This table reports order-level regressions of the cash indicator on an indicator for the payday window and the utilization bins of equation (1). The payday window runs from the 15th to the 17th and from the last day of the month to the 2nd of the next month. Column 1 enters the payday window alone, column 2 adds the utilization bins with $[0, 20)$ omitted, and column 3 interacts the bins with the window. The two p-values are from tests that the four interactions are jointly zero and that the top-bin interaction is zero. The sample is the regression sample of Section 2, excluding orders with zero value or with day-before utilization above 105 percent. All columns include user, bill-month, and category fixed effects. Controls are the log of order value, the log of RappiCard purchases in the previous billing cycle, and an indicator for no purchases in that cycle. Standard errors clustered by user in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: First stage: acceptance at the cutoff

	Accepts an increase			
	pp	t	h	Eff. N
Pooled	+18.5	15.1	12.1	7,722
Utilization < 0.6	+30.0	13.9	15.1	3,912
Utilization ≥ 0.6	+12.3	10.8	16.0	5,807
Non-revolver	+25.2	13.5	13.3	4,592
Revolver	+12.0	6.9	5.2	2,121

Notes. This table reports the change at the credit score cutoff in the share of households that accept a credit limit increase, in percentage points. The sample is the user-by-campaign panel of the four campaigns described in Section 2, with scores centered at each campaign's cutoff, restricted to campaigns within 335 days of card approval so that Rappi orders are observed in the 30 days after the campaign. This is the sample of Table 6. Rows split households by utilization before the campaign and by revolving status before the campaign. Estimates are by RDHonest with campaign fixed effects and a triangular kernel. The utilization rows use the bandwidth of the matching row of Table 6; the revolving-status rows use the bandwidth chosen by the estimator. h is the bandwidth, in score points, and Effective N is the number of observations within it. t is the estimate over its standard error, clustered by user.

Table 6: Reduced-form results by utilization and revolving status

	Jump	95% CI	h	Eff. N
Panel A. By pre-CLI utilization				
All	-0.133^{**}	$[-0.247, -0.019]$	12.1	7,642
Utilization < 0.6	-0.058	$[-0.243, +0.127]$	15.1	3,838
Utilization ≥ 0.6	-0.140^{**}	$[-0.261, -0.019]$	16.0	5,772
Difference	-0.082	$p = 0.39$		
Panel B. By pre-period revolver status				
All utilization				
Non-revolver	-0.128^{*}	$[-0.280, +0.024]$	14.2	4,333
Revolver	-0.144^{*}	$[-0.307, +0.018]$	11.0	3,650
Difference	-0.016	$p = 0.87$		
Utilization ≥ 0.6				
Non-revolver	-0.255^{*}	$[-0.530, +0.020]$	13.3	1,861
Revolver	-0.100	$[-0.235, +0.034]$	13.9	3,361
Difference	$+0.154$	$p = 0.29$		

Notes. This table reports the change at the credit score cutoff in the number of cash-paid Rappi orders in the 30 days after the campaign, which is zero for households with no order. Panel A splits households by utilization before the campaign and Panel B by revolving status before the campaign. Estimates are by RDHonest with campaign fixed effects and a triangular kernel, on the sample of Table 5. h is the bandwidth chosen by the estimator, in score points, and Effective N the number of observations within it. Brackets are honest 95 percent confidence intervals from standard errors clustered by user. The difference rows report the high-utilization minus the low-utilization jump (Panel A) and the revolver minus the non-revolver jump (Panel B), with the p-value of a test of equal jumps that treats the two cells as independent. ** marks an honest 95 percent interval that excludes zero and * an honest 90 percent interval that excludes zero.

Table 7: Measured inputs and estimated parameters

Panel A. Measured and imported					
Utilization (percent)	$[0, 20)$	$[20, 40)$	$[40, 60)$	$[60, 80)$	$[80, +)$
$\kappa^R(u)$, revolving (cents per dollar)	25.3	31.4	29.8	29.5	25.5
Pr(non-revolving cycle turns revolving)	0.086	0.150	0.183	0.213	0.305
$\kappa^N(u)$, non-revolving (cents per dollar)	2.2	4.7	5.4	6.3	7.8
p, monthly frequency of a measured shock	0.038				
θ_m, credit-financed share of a measured shock	0.207				
G, shock size	truncated exponential; mean 3,000, cap 30,000 pesos				
Panel B. Estimated					
$\mathbb{E}[\theta]\,\rho = (p\,\theta_m + q)\,\rho$	0.434 [0.312, 0.510]				
F shape (Weibull)	2.70 [2.20, 3.39]				
F scale (Weibull)	1.93 [1.48, 2.50]				
π_0, orders paid in cash at any price	0.0739 [0.0725, 0.0752]				
J statistic, 4 degrees of freedom (p-value)	9.80 (0.04)				

Notes. This table reports the measured inputs (Panel A) and the estimated parameters (Panel B) of the model of Section 6. In Panel A, $\kappa^R(u)$ is the interest accrued over its life by a dollar entering revolving debt in the first 12 months after card approval, averaged by utilization bin at origination; $\kappa^N(u)$ is $\kappa^R(u)$ times the probability that a non-revolving cycle turns revolving at the next statement. p is the monthly probability of a RappiCard transaction at a hotel, car rental firm, private hospital or travel agency; θ_m is the average hold placed by these merchants in a month with a hold, relative to the mean shock size. G is calibrated to ENIGH 2022 shock-prone expenditure per adult per month. Panel B reports classical minimum distance estimates on ten cash shares (five utilization bins by revolving status) with five free parameters (π_0, the shape and scale of F, q and ρ), minimizing the sum of squared differences between the model and data cash shares, each divided by the standard error of the data cash share. Brackets are 95 percent intervals from 500 bootstrap draws that resample the moments and the measured interest prices. q and ρ are reported in Appendix B.3.

Table 8: The cost of credit on cash-paid orders

			Cost of credit if charged (cents per dollar)		
	Cash orders	Value (million pesos)	Interest	Shadow cost	Total
Non-revolving	54,526	12.57	4.73	16.29	21.02
Revolving	47,823	11.07	27.20	23.70	50.90
All	102,349	23.64	15.25	19.76	35.01

Notes. This table reports the cash-paid Rappi orders of the estimation sample over the 12 months after card approval, by the revolving status of the billing month, their value, and the cost of credit per dollar each would have carried if charged. Each order is priced at $\text{TMCC}^I(u)$ of equation (6) at the estimates in Table 7, with u the household's utilization on the day before the order, split into the interest price and the shadow cost. The cost is averaged over the cash orders in each row, weighted by order value. Orders that would not have fit under 105 percent of the limit on the day before the order are excluded (8,541 of 110,890 cash orders, 3.39 million of 27.04 million pesos, almost all above 80 percent utilization). A household can be revolving in some months and non-revolving in others: 25,366 households paid cash at least once, 16,074 of them in non-revolving months and 14,537 in revolving months.

Table 9: Change in the cost of credit under limit and interest rate changes

	Non-revolving				Revolving			
Utilization (percent)	Baseline TMCC	Limit +10%	Limit −10%	Rate −10%	Baseline TMCC	Limit +10%	Limit −10%	Rate −10%
$[0, 20)$	6.35	−0.69	0.85	−0.21	28.36	−0.54	0.67	−2.50
		$[-0.83, -0.44]$	$[0.55, 1.02]$	$[-0.22, -0.21]$		$[-0.63, -0.36]$	$[0.45, 0.78]$	$[-2.52, -2.46]$
$[20, 40)$	12.44	−1.34	1.68	−0.46	36.78	−0.99	1.28	−3.08
		$[-1.59, -0.88]$	$[1.12, 1.99]$	$[-0.47, -0.44]$		$[-1.16, -0.68]$	$[0.89, 1.49]$	$[-3.12, -2.99]$
$[40, 60)$	17.95	−2.29	2.97	−0.52	39.47	−1.88	2.52	−2.89
		$[-2.69, -1.58]$	$[2.15, 3.46]$	$[-0.54, -0.49]$		$[-2.18, -1.33]$	$[1.84, 2.92]$	$[-2.95, -2.74]$
$[60, 80)$	26.54	−4.10	5.89	−0.59	46.05	−3.66	5.49	−2.78
		$[-4.74, -3.17]$	$[5.09, 6.70]$	$[-0.61, -0.53]$		$[-4.18, -2.82]$	$[4.58, 6.25]$	$[-2.89, -2.54]$
$[80, +)$	50.60	−11.59	6.90	−0.68	68.46	−11.01	4.70	−2.19
		$[-15.12, -9.38]$	$[5.38, 9.48]$	$[-0.73, -0.55]$		$[-14.90, -8.27]$	$[3.50, 6.44]$	$[-2.39, -1.76]$

Notes. This table reports the change in $\text{TMCC}^I(u)$ of equation (6), in cents per dollar charged to the card, when every household's credit limit rises by 10 percent, falls by 10 percent, or its contracted interest rate falls by 10 percent (0.62 percentage points per month at the median rate), for non-revolving and revolving months. The baseline columns report $\text{TMCC}^I(u)$ before the change, as point estimates. Negative values are a lower cost of credit. Prices are evaluated at the estimates in Table 7, holding the interest price, the probability that a non-revolving cycle turns revolving, ρ, and the length of revolving spells at their values before the change. The unit is a user-month with purchases on RappiCard: 554,971 months of 59,752 households in the estimation sample, excluding months in which the account is blocked. Utilization is measured at the statement, and each row averages over the months in the bin with equal weight. Brackets are 95 percent intervals from 500 bootstrap draws of the estimated parameters.

A Model Proofs

Throughout this appendix, we denote $D_1 = D_0 + (1-\pi)x$ at the chosen π and $H = \bar{B} - D_1$. Differentiation is with respect to D_0 at fixed π, so $\partial D_1/\partial D_0 = 1$ and $\partial H/\partial D_0 = -1$; $\bar{B}$ is fixed, so convexity in D_0 is convexity in utilization $D_0/\bar{B}$. We write $\mathbb{E}_y[\cdot]$ for the expectation over y at a given ξ, and c_2^* for consumption when the shock equals H, which is the same in both regimes by continuity. Under risk neutrality every sign below uses only $\rho > r_{\text{eff}}$ and, for convexity, a decreasing shock density. With risk aversion the derivatives gain terms in $-u''$ and u''', which are positive under CRRA for any $\gamma > 0$, so each result extends to every $\gamma \geq 0$.

We set $\theta = 1$. For a realization $\theta \in (0,1)$, each proof goes through with the regime boundary $\xi = H/\theta$ in place of $\xi = H$, and no sign changes. The household does not know θ when it pays at $t = 0$, so its cost of credit is the average of these costs over the values θ can take. Each positive value gives an increasing and convex cost, and a draw of $\theta = 0$ gives a flat one, so the average is increasing and convex whenever $\mathbb{E}[\theta] > 0$. The gap of Proposition 3 is positive at every θ and therefore on average.

Proposition 1 (Monotonicity)

Proof. Write the two terms of (2) as integrals over the shock density,

$$\text{TMCC} = \beta r_{\text{eff}} \int_0^H \mathbb{E}_y[u'(c_2^a)]\, g(\xi)\, d\xi + \beta\rho \int_H^{\bar{\xi}} \mathbb{E}_y[u'(c_2^o)]\, g(\xi)\, d\xi,$$

and apply the Leibniz rule to each, using $\partial c_2^a/\partial D_0 = -(1+r_{\text{eff}})$ and $\partial c_2^o/\partial D_0 = -(1+\rho)$. The boundary terms of the two integrals carry the same $\mathbb{E}_y[u'(c_2^*)]$, because c_2 is continuous at $\xi = H$, and they add up to one term:

$$\frac{\partial \text{TMCC}}{\partial D_0} = \underbrace{\beta(\rho - r_{\text{eff}})\, \mathbb{E}_y[u'(c_2^*)]\, g(H)}_{\text{(i)}} + \underbrace{\beta r_{\text{eff}}(1+r_{\text{eff}}) \int_0^H \mathbb{E}_y[-u''(c_2^a)]\, g\, d\xi}_{\text{(ii)}}$$
$$+ \underbrace{\beta\rho(1+\rho) \int_H^{\bar{\xi}} \mathbb{E}_y[-u''(c_2^o)]\, g\, d\xi}_{\text{(iii)}}.$$

Term (i) is positive because $\rho > r_{\text{eff}}$ and $u' > 0$. Terms (ii) and (iii) are positive because $-u'' > 0$ under CRRA with $\gamma > 0$, and vanish at $\gamma = 0$, where the derivative is term (i) alone. Hence $\partial \text{TMCC}/\partial D_0 > 0$. □

Term (i) is the boundary term: the added overflow mass $g(H)$ from one more dollar, priced at the premium $\rho - r_{\text{eff}}$. Its sign uses only $\rho > r_{\text{eff}}$ and $u' > 0$, so it survives the risk-neutral limit $\gamma \to 0$. This decomposition supports the claim in Section 4.2 that the mechanism prices credit capacity and not wealth.

Proposition 2 (Convexity)

Proof. Differentiate the expression for $\partial \text{TMCC}/\partial D_0$ obtained in the proof of Proposition 1 once more. The simplification is that consumption at the regime boundary

$$c_2^* = y + M_1 - (1 + r_{\text{eff}})(D_1 + H) = y + M_1 - (1 + r_{\text{eff}})\bar{B},$$

does not depend on D_0, since $D_1 + H = \bar{B}$ identically. Term (i) therefore differentiates through g alone. Collecting the four resulting terms,

$$\frac{\partial^2 \text{TMCC}}{\partial D_0^2} = \underbrace{-\beta(\rho - r_{\text{eff}})\,\mathbb{E}_y[u'(c_2^*)]\,g'(H)}_{\text{(a)}} + \underbrace{\beta\big[\rho(1+\rho) - r_{\text{eff}}(1+r_{\text{eff}})\big]\,\mathbb{E}_y[-u''(c_2^*)]\,g(H)}_{\text{(b)}}$$
$$+ \underbrace{\beta r_{\text{eff}}(1+r_{\text{eff}})^2 \int_0^H \mathbb{E}_y[u'''(c_2^a)]\,g\,d\xi}_{\text{(c)}} + \underbrace{\beta\rho(1+\rho)^2 \int_H^{\bar{\xi}} \mathbb{E}_y[u'''(c_2^o)]\,g\,d\xi}_{\text{(d)}}.$$

Term (a) is positive whenever $g'(H) < 0$. Term (b) collects the two boundary terms of the Leibniz rule and is positive because $\rho(1+\rho) > r_{\text{eff}}(1+r_{\text{eff}})$, which follows from $\rho > r_{\text{eff}} \geq 0$, and $-u'' > 0$. Terms (c) and (d) are positive because $u''' > 0$ under CRRA. Terms (b), (c) and (d) vanish at $\gamma = 0$, where the second derivative is term (a) alone. Hence $\partial^2 \text{TMCC}/\partial D_0^2 > 0$. □

A decreasing shock density is therefore sufficient for convexity at every degree of risk aversion, and it is the only restriction on G the proof uses. Under risk neutrality the second derivative is $-\beta(\rho - r_{\text{eff}})g'(H)$, so a decreasing density is then necessary as well. For the untruncated exponential, $1 - G(H) = e^{-\lambda H}$, the risk-neutral cost is $\beta r_{\text{eff}} + \beta(\rho - r_{\text{eff}})e^{-\lambda H}$

as in (3), and its second derivative in D_0 is $\beta(\rho - r_{\text{eff}})\lambda^2 e^{-\lambda H} > 0$. The truncation of Assumption 1(i) adds a constant and a scale factor to $1-G(H)$, so the derivatives keep their signs.

Proposition 3 (Revolving gap)

Proof. For a revolving household $r_{\text{eff}} = r > 0$ and for a non-revolving household $r_{\text{eff}} = 0$, so the non-revolving interest channel is zero while the revolving interest channel is strictly positive. For the shadow cost, overflow-regime consumption satisfies $c_2^{o,R} = c_2^{o,N} - r\bar{B}$, and u' is decreasing, so $u'(c_2^{o,R}) > u'(c_2^{o,N})$ everywhere on the overflow region. The revolving household's shadow cost therefore exceeds the non-revolving household's, and $\text{TMCC}_R - \text{TMCC}_N > 0$ at every D_0. □

Write

$$\text{TMCC}_R - \text{TMCC}_N = \beta r\, \mathbb{E}\Big[u'(c_2^{a,R})\mathbf{1}\{\xi \le H\}\Big] + \beta\rho\, \mathbb{E}\Big[\Big(u'(c_2^{o,R}) - u'(c_2^{o,N})\Big)\,\mathbf{1}\{\xi > H\}\Big].$$

As D_0 rises, H falls and $P(\xi > H)$ rises, so the second component increases: its integrand equals $r\bar{B}\cdot(-u''(\tilde{c})) > 0$ for some intermediate $\tilde{c}$ and is itself increasing in D_0. The first component is the interest on dollars that stay on the card, so it counts only the absorbed region; it differs from the interest term of Section 4.3, which charges r_{eff} on every dollar. It falls toward zero as $D_0 \to \bar{B}$, because the absorbed region empties. Under risk neutrality it equals $\beta r G(H)$ and falls monotonically. As $D_0 \to \bar{B}$ the difference converges to $\beta\rho\,\mathbb{E}[u'(c_2^{o,N} - r\bar{B}) - u'(c_2^{o,N})] > 0$, a positive finite limit. The difference is therefore positive throughout and contracts near the limit, where the interest channel loses its mass.

Free substitution

The claim in Section 4.2 is that the mechanism disappears when cash and available credit are interchangeable for meeting the $t = 1$ shock. Free substitution means the household funds the shock using cash or credit, and turns to the outside source only once both are exhausted. Two facts follow.

First, the resources available before the outside source do not depend on the payment choice. Substituting $D_1 = D_0 + (1-\pi)x$ and $M_1 = M_0 - \pi x$, the terms in πx cancel and $H + M_1 = M_0 + \bar{B} - D_0 - x$ for every $\pi \in [0,1]$. The two cheaper sources fund $\min(\xi,\ M_1 + H)$ and the residual $(\xi - M_1 - H)^+$ goes to the outside source, so the outside draw is the same for every realization of ξ and every π.

Second, the shadow cost is identically zero at every D_0, and $\partial \mathrm{TMCC}/\partial \bar{B} = 0$ for every $\gamma \geq 0$. Under free substitution, the shock is met from cash when $\xi \leq M_1$, from cash and then the card when $M_1 < \xi \leq M_1 + H$, and from the outside source when $\xi > M_1 + H$. Differentiate $\mathbb{E}[u(c_2)]$ in π over the three regions. In the first, $\partial c_2/\partial \pi = -x + (1 + r_{\text{eff}})x = r_{\text{eff}} x$. In the second, the two movements offset and $\partial c_2/\partial \pi = 0$. In the third, the outside draw is invariant by the first fact, so again $\partial c_2/\partial \pi = 0$. The boundary between the first two regions moves with π but c_2 is continuous there, and the boundary with the third region does not move at all. Collecting terms leaves only the interest saving over $\xi \leq M_1$. Neither that integrand nor its upper limit involves $\bar{B}$, so the derivative in $\bar{B}$ is zero. At $\gamma = 0$ the surviving term is $\beta r_{\text{eff}} G(M_1)$, which does not vary with D_0.

B Structural Estimation

This appendix defines the measured inputs, the estimator, and the checks behind the estimates of Section 6.

B.1 Measured inputs

Revolving spells We measure the interest price from the monthly RappiCard statements of the households in the estimation sample. A revolving spell is a run of consecutive statements with a positive revolving balance. The revolving balance excludes interest-free and interest-bearing installment plans. We also remove the payments that retire those plans, so only payments toward the revolving balance count. A spell ends when the revolving balance reaches zero or when the panel ends. We classify a spell as written off once the account falls four payments behind: 95 percent of the accounts that reach this state are charged off, and fewer than 1 percent return to good standing.

Following a dollar of debt We follow the revolving balance from statement to statement. The new revolving debt of month m, n_m, is the rise in the balance beyond the interest charged, floored at zero. We label each month's new debt by the household's utilization at the previous statement and follow it separately. Repayment is pro rata: every dollar carried from the previous statement shrinks by the same factor

$$s_m \;=\; \frac{b_m - n_m}{b_{m-1}}, \tag{8}$$

where b_m is the revolving balance on statement m. We cap s_m at one plus the interest charged on statement m divided by b_{m-1}, so late fees never enter the interest price. In months when the account is behind on its payments, we floor s_m at one minus the payment divided by b_{m-1}, so a settlement or a write-down does not count as repayment. The bank charges interest on the closing balance of the same statement, so the interest rate of month m, r_m, is the interest charged on statement m divided by b_m. A dollar of debt originated in month t is still owed in month m in the proportion $\prod_{k=t+1}^{m} s_k$, and its interest price is

$$\kappa(t) \;=\; \sum_{m=t}^{W} r_m \prod_{k=t+1}^{m} s_k, \tag{9}$$

where W is the last month of the spell in which the household makes a payment. Stopping at W leaves out interest that is booked after the household stops paying and is never collected. We do not discount. $\kappa^R(u)$ averages $\kappa(t)$ over the dollars of new debt originated in utilization bin u. The duration of the debt is the same sum with r_m set to one, the expected number of months the dollar stays in debt.

Non-revolving cycles The probability that a non-revolving cycle turns revolving is the share of non-revolving statements in a utilization bin that are followed by a revolving statement. We measure it on consecutive statements of the same household. A statement enters if the next statement is the following month, the account is not blocked, and utilization at the statement is below 105 percent. The share within each of the five utilization bins gives the five probabilities.

Measured shock We measure the credit-financed shock from RappiCard transactions at hotels, car rental firms, private hospitals and travel agencies. These merchants typically require a credit pre-authorization. A hold is an authorized charge that is never settled or is settled for less than the authorized amount. We drop authorizations that match a settlement at the same merchant within seven days and within 2 percent of the amount. We also drop hospital charges below 500 pesos, made at a public hospital, or paid through a payment processor. We drop transactions in months when the account is blocked or over its limit. An exposure month is a billing month in the first twelve months after card approval in which the account is neither blocked nor over its limit. p is the share of exposure months with a transaction at these merchants, pooled over the households in the estimation sample; we count every transaction, since not every pre-authorization appears on the statement. θ_m is the average total hold amount in a household-month with a hold, 620 pesos, divided by the mean shock size of 3,000 pesos. The measured shock is likely to be a lower bound, since households at high utilization are less likely to put pre-authorizations on this card.

B.2 Estimation and inference

Objective Let $\hat{\pi}_j$ be the ten cash shares, five utilization bins by revolving status, and σ_j their standard errors. We choose π_0, the shape and scale of F, q and ρ to minimize

$$\sum_{j=1}^{10} \frac{\big(\hat{\pi}_j - \pi_j(\pi_0, F, q, \rho)\big)^2}{\sigma_j^2}, \tag{10}$$

where π_j is the model cash share (7) averaged over the orders in cell j. Each order is evaluated at its own available credit. The data cash shares and their standard errors come from the cash-share regression in column 3 of Table 2. The covariance between moments does not enter the weights. We minimize using bounded nonlinear least squares from several starting values, with π_0 and q between 0 and 0.5.

Overidentification The overidentification test does not use this objective. It refits the model with the full covariance of the ten cash shares as the weight matrix. The fixed effects leave that covariance with rank nine, so we use its generalized inverse. J is the minimized

objective of this refit. It is distributed χ^2 with four degrees of freedom, nine minus the five parameters.

Inference The intervals are percentile intervals from 500 bootstrap draws. Each draw adds noise to the ten cash shares from their estimated covariance. It also recomputes $\kappa^R(u)$ and the probability that a non-revolving cycle turns revolving on a sample of households drawn with replacement. p, θ_m, G and the available credit of each order stay fixed. We re-estimate in every draw and compute $\mathbb{E}[\theta]\rho$ draw by draw.

B.3 The unmeasured shock and the overflow price

The cash shares pin the product $\mathbb{E}[\theta]\rho$, but not q and ρ separately. A more frequent unmeasured shock with a lower overflow price produces nearly the same cost of credit at every utilization level, so the two trade off along a ridge. The estimates are $\rho = 3.32$, with a 95 percent interval [1.47, 7.05], and $q = 0.123$, with interval [0.039, 0.313]. Only their product enters the cost of credit. Because p and θ_m are fixed, the interval of $\mathbb{E}[\theta]$ is the interval of q shifted by $p\,\theta_m$.

B.4 Recovery from simulated data

We simulate 1,000 sets of the ten cash shares from the model at the estimates. Each set adds noise drawn from the estimated covariance of the cash shares. We re-estimate on each set, with the measured inputs at their point values. The median estimate of $\mathbb{E}[\theta]\rho$ is 0.430, against the true value of 0.434, and 95 percent of the estimates lie between 0.31 and 0.50. The estimates of q and ρ spread along the ridge of Appendix B.3, so we compare medians. They are 0.123 against 0.123 for q, and 3.27 against 3.32 for ρ.

B.5 Robustness

Table D13 reports $\mathbb{E}[\theta]\rho$ under each alternative. First, we re-estimate the model with a lognormal and a log-logistic F, two other distributions that also have two parameters and can be convex over the range of costs in the data. A lognormal or log-logistic F gives $\mathbb{E}[\theta]\rho$

of 0.425 and 0.433, close to the headline fitted with a Weibull F. The shadow cost of a non-revolving household is 5 cents at 20 percent utilization and 24 cents at 80 percent under a lognormal or log-logistic F, as under the Weibull.

Second, we test how our estimates change when the mean shock size changes. Halving the mean to 1,500 pesos lowers the shadow cost of a non-revolving household from 5 to 1 cent at 20 percent utilization and from 24 to 15 cents at 80 percent. Doubling it to 6,000 pesos raises the shadow cost to 10 and 30 cents. Of the three means, 3,000 pesos fits the cash shares best, and 1,500 pesos fits much worse. Moreover, the headline holds the shock size fixed across credit limits. If it instead rises with the household's total credit limit, with elasticity 0.14, the product is 0.438.[31]

We also fit two disjoint groups separately: households for which RappiCard accounts for at least 70 percent of their credit card limits at card approval, and the rest. $\mathbb{E}[\theta]\rho$ is 0.419 and 0.440, well within the bootstrap interval of the headline estimate and not statistically different from each other. Finally, we measure the interest inputs by utilization bin over the whole panel instead of the first 12 months after card approval. $\kappa^R(u)$ is then lower and flatter, and the product is 0.357.

[31]The elasticity is the product of two elasticities: of shock-prone expenditure to income in ENIGH, 0.467, and of income to the total credit limit, 0.295.

C Additional figures

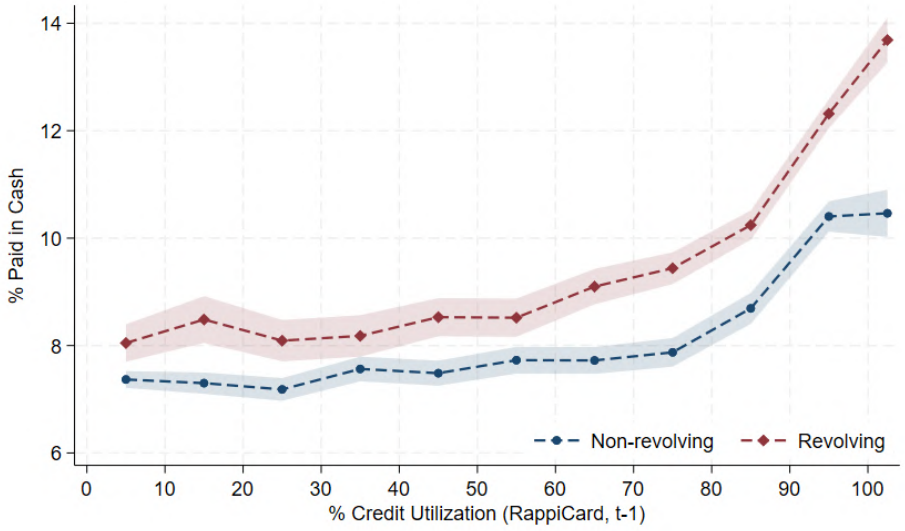


Figure C1: Fitted cash share in finer utilization bins

Notes. This figure plots the fitted share of orders paid in cash, in percent, from the specification in column 3 of Table 2 with eleven utilization bins in place of five: ten-point bins up to 80 percent, then [80,90), [90,100), and [100,105], separately for revolving and non-revolving months. Points are plotted at bin midpoints, and the [100,105] bin at 102.5. Each point is the average of the regression's fitted values over the orders in the bin and revolving state. Shaded bands are 95 percent confidence intervals from standard errors clustered by user.

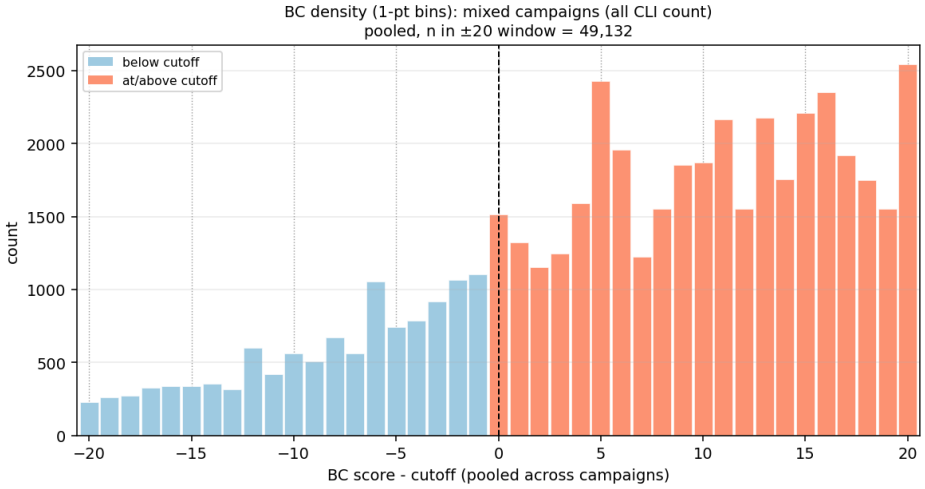
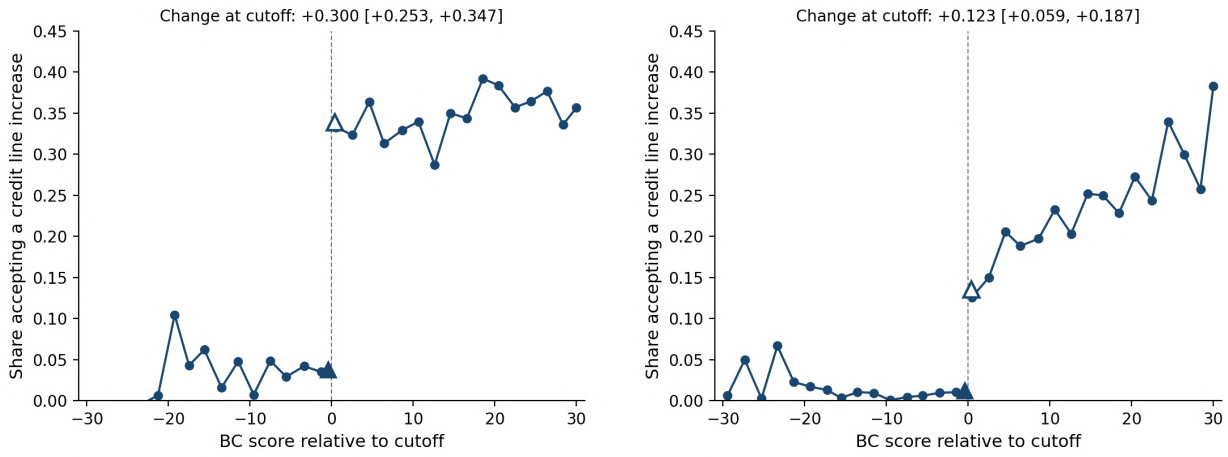


Figure C2: Fine-bandwidth density of credit scores

Notes. This figure plots the number of households at each credit score, centered at each campaign's cutoff and pooled across the four campaigns, within 20 points of the cutoff, on the sample of Table 5. Each bar is one score point.

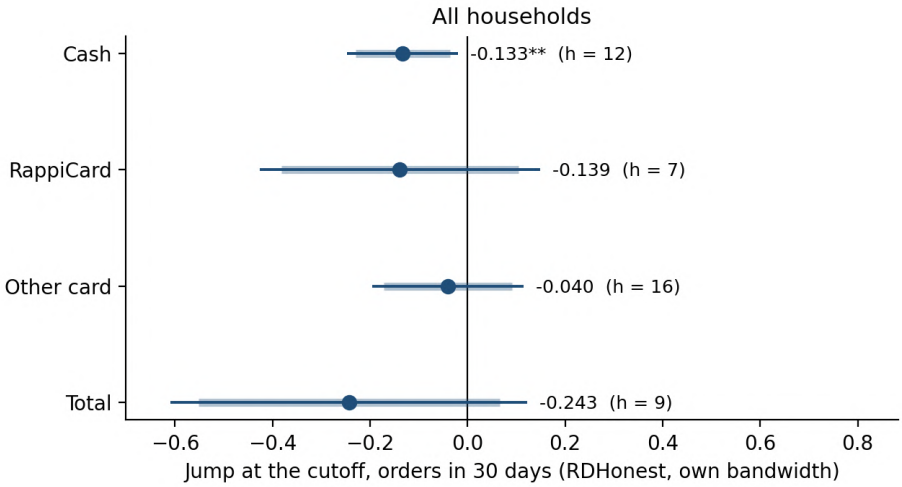


(a) Utilization below 60 percent

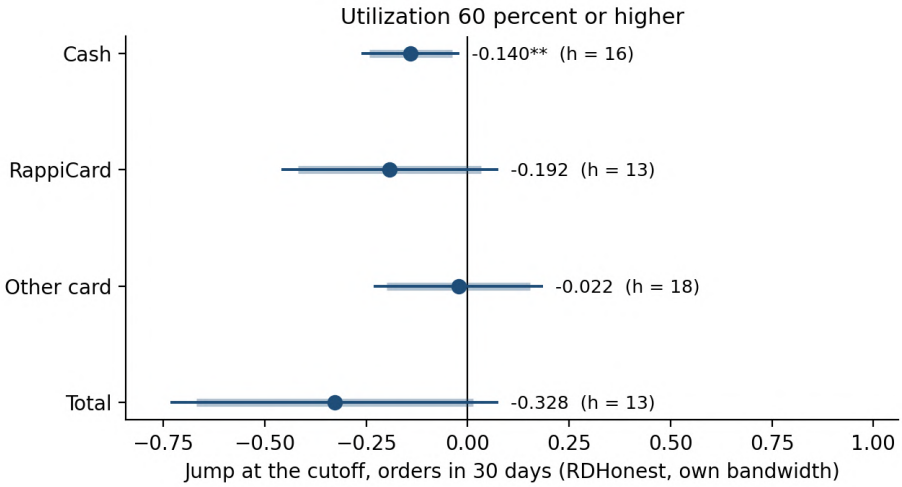
(b) Utilization at or above 60 percent

Figure C3: First stage by pre-campaign utilization

Notes. This figure plots Figure 6 separately for households below (left) and at or above (right) 60 percent utilization before the campaign. Circles are means in two-point bins within 30 points of the cutoff. Triangles are the local linear predictions at the cutoff from below (solid) and from above (hollow), with a triangular kernel and each group's bandwidth from Table 5, 15.1 points for low and 16.0 points for high utilization; their difference is the group's estimate in Table 5. Both panels share a vertical scale.



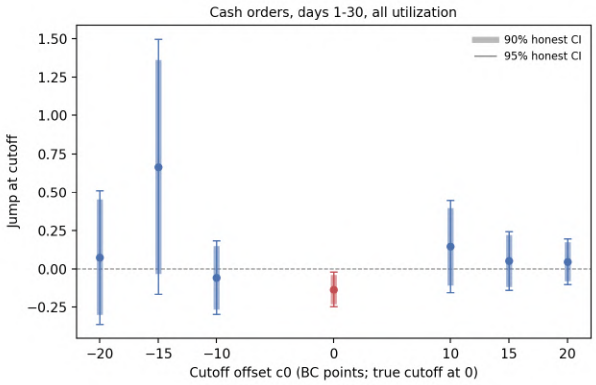
(a) All users



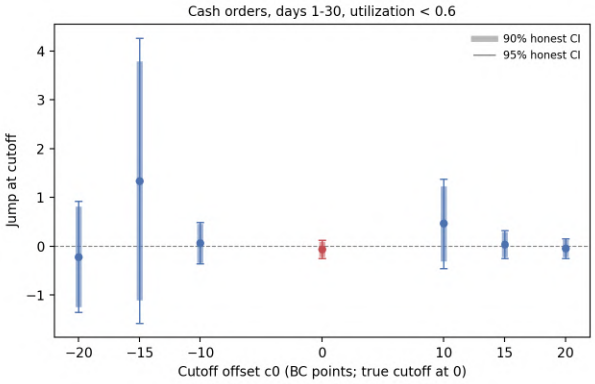
(b) Utilization at or above 60 percent

Figure C4: Reduced form by payment instrument

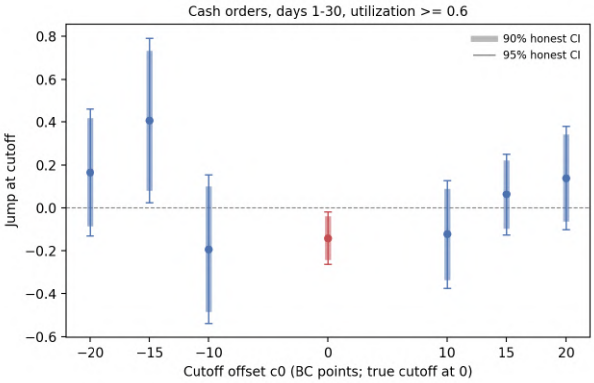
Notes. This figure plots the reduced-form jump at the score cutoff in the number of Rappi orders in the 30 days after the campaign date, by payment instrument, for all households in Panel A and for households with pre-campaign utilization of 60 percent or higher in Panel B, in the four campaigns of Table 6. Each estimate is RDHonest with a triangular kernel, campaign fixed effects, standard errors clustered by user, and its own bandwidth chosen by the estimator (in parentheses). Thin lines are 95 percent honest intervals; thick lines are 90 percent intervals. The cash estimates are the corresponding estimates of Table 6.



(a) All users



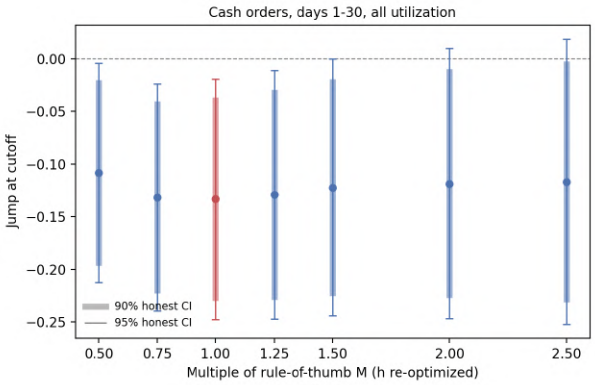
(b) Utilization below 60 percent



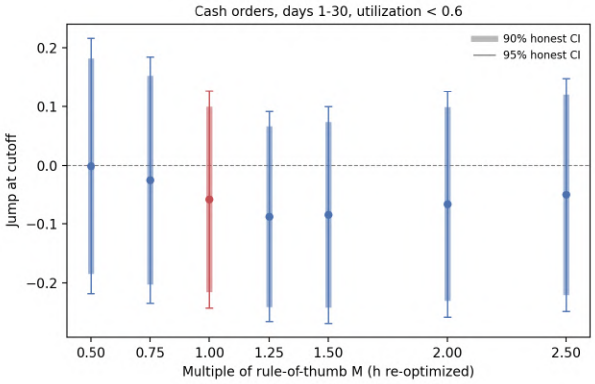
(c) Utilization at or above 60 percent

Figure C5: Placebo cutoffs

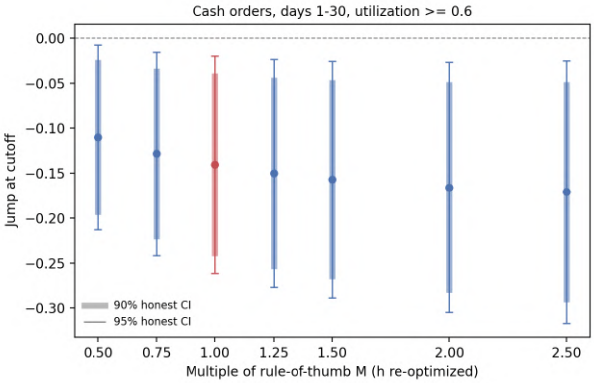
Notes. This figure plots the change in the count of cash-paid Rappi orders in the 30 days after the campaign at the true cutoff (red) and at false cutoffs offset by 10, 15, and 20 score points on either side (blue), for all households and separately by utilization before the campaign. Each false cutoff is estimated on one side of the true cutoff only, using scores below it for negative offsets and scores at or above it for positive offsets. The outcome is residualized on campaign fixed effects within that side, and RDHonest selects M and the bandwidth separately for each window. Thin capped lines are honest 95 percent confidence intervals and thick bars honest 90 percent intervals, from standard errors clustered by user.



(a) All users



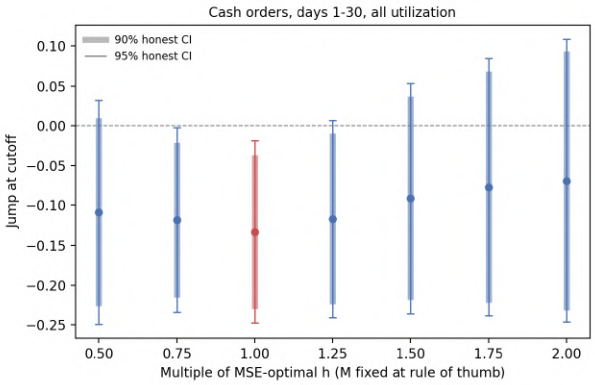
(b) Utilization below 60 percent



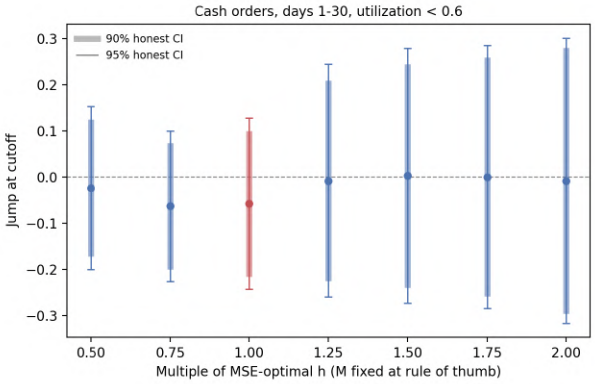
(c) Utilization at or above 60 percent

Figure C6: Robustness to the curvature bound M

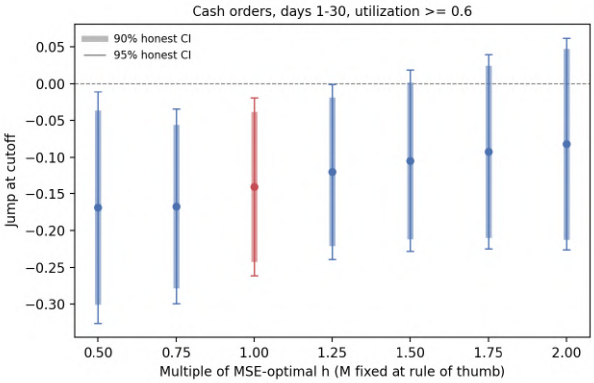
Notes. This figure plots the change in the count of cash-paid Rappi orders in the 30 days after the campaign with the bound M on the second derivative of the conditional mean set to 0.5 to 2.5 times the estimator's rule-of-thumb value, with the bandwidth re-selected at each M, for all households and separately by utilization before the campaign. The red point is the estimate of Table 6. Estimates are by RDHonest with campaign fixed effects and a triangular kernel. Thin capped lines are honest 95 percent confidence intervals and thick bars honest 90 percent intervals, from standard errors clustered by user.



(a) All users



(b) Utilization below 60 percent



(c) Utilization at or above 60 percent

Figure C7: Robustness to the bandwidth

Notes. This figure plots the change in the count of cash-paid Rappi orders in the 30 days after the campaign with the bandwidth set to 0.5 to 2 times the estimator's choice and M held at its rule-of-thumb value, for all households and separately by utilization before the campaign. The red point is the estimate of Table 6. Estimates are by RDHonest with campaign fixed effects and a triangular kernel. Thin capped lines are honest 95 percent confidence intervals and thick bars honest 90 percent intervals, from standard errors clustered by user.

D Additional tables

Table D1: Cash use by purchase category, revolving status, and utilization

Panel A. Share of orders paid in cash (percent)			
	Non-revolving	Revolving	Difference (pp)
Grocery	6.7	11.7	5.02***
Restaurants	8.3	12.3	3.95***
Temptation goods	5.8	8.8	2.96***
Others	6.3	9.4	3.15***
All	7.9	11.8	3.86***
Panel B. Median utilization on the day before the order (percent)			
	Card-paid orders	Cash-paid orders	Difference (pp)
Non-revolving months	30.9	46.4	15.5***
Revolving months	77.5	85.8	8.2***

Notes. This table reports cash use by purchase category and revolving status at the order level. Revolving months are billing months in which the account carries an interest-bearing balance. Panel A reports the share of orders paid in cash, in percent, by purchase category: grocery covers supermarket, pharmacy, and convenience stores, and temptation goods are orders containing alcohol or tobacco. The difference is the revolving share minus the non-revolving share, with stars from a regression of the cash indicator on the revolving indicator within the category, standard errors clustered by user. Panel B reports the median RappiCard utilization on the day before the order, in percent, at card-paid orders, which include every non-cash instrument, and at cash-paid orders. The difference is cash minus card, with stars from a median regression of utilization on the cash indicator within the revolving state, robust standard errors. The sample is the regression sample of Section 2; Panel B excludes orders with day-before utilization above 105 percent. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D2: Cash use and aggregate credit capacity: alternative revolving measure

	Indicator for cash payment			
	(1)	(2)	(3)	(4)
Aggregate credit utilization, % (omitted bin [0,20))				
[20,40)	0.0003	0.0002	-0.0001	-0.0020
	(0.0025)	(0.0025)	(0.0026)	(0.0028)
[40,60)	0.0013	0.0016	0.0013	-0.0017
	(0.0030)	(0.0030)	(0.0030)	(0.0034)
[60,80)	0.0046	0.0046	0.0045	0.0043
	(0.0034)	(0.0034)	(0.0034)	(0.0036)
[80,+)	0.0161***	0.0166***	0.0163***	0.0176***
	(0.0033)	(0.0033)	(0.0034)	(0.0038)
Revolving on any card	0.0048	0.0047	0.0046	0.0037
	(0.0030)	(0.0031)	(0.0031)	(0.0032)
[20,40) × Revolving on any card	-0.0020	-0.0020	-0.0018	-0.0009
	(0.0034)	(0.0034)	(0.0034)	(0.0036)
[40,60) × Revolving on any card	0.0029	0.0023	0.0027	0.0042
	(0.0039)	(0.0039)	(0.0039)	(0.0042)
[60,80) × Revolving on any card	0.0057	0.0057	0.0058	0.0042
	(0.0042)	(0.0042)	(0.0042)	(0.0044)
[80,+) × Revolving on any card	0.0141***	0.0127***	0.0131***	0.0083*
	(0.0043)	(0.0043)	(0.0043)	(0.0045)
Controls	No	Yes	Yes	Yes
Bureau controls	No	No	No	Yes
Category FE	No	No	Yes	Yes
Cycle-position FE	No	No	Yes	Yes
N	643,974	637,097	634,676	593,803
R^2	0.429	0.428	0.428	0.425

Notes. This table reports estimates of equation (1) with bins defined on aggregate utilization across all of the household's credit cards, as in Table 3, and the revolving indicator widened to any card. The bureau does not flag whether another card's balance is interest-bearing, so another card is classified as revolving when its payment is less than half its outstanding balance. Revolving on any card equals one if the RappiCard account carried an interest-bearing balance at the most recent statement before the order, or if any other card is revolving on a bureau report no older than one month. The sample is that of Table 3. All columns include user and bill-month fixed effects. Controls, bureau controls, and cycle-position fixed effects are those of Table 3. Standard errors clustered by user in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D3: Cash use and the interaction of the two utilizations

	Indicator for cash payment			
	(1) Linear	(2) Linear	(3) 3×3 bins	(4) 3×3 bins
Own-card utilization	0.0178***	0.0177***		
	(0.0040)	(0.0040)		
Other-card utilization	-0.0040	-0.0040		
	(0.0048)	(0.0048)		
Own × Other utilization	0.0286***	0.0286***		
	(0.0065)	(0.0065)		
Own-card utilization [40,80)			0.0060***	0.0060***
			(0.0020)	(0.0020)
Own-card utilization [80,+)			0.0207***	0.0205***
			(0.0029)	(0.0029)
Other-card utilization [40,80)			0.0011	0.0012
			(0.0023)	(0.0023)
Other-card utilization [80,+)			0.0067**	0.0067**
			(0.0033)	(0.0033)
Own [40,80) × Other [40,80)			0.0010	0.0010
			(0.0027)	(0.0027)
Own [40,80) × Other [80,+)			0.0014	0.0013
			(0.0036)	(0.0036)
Own [80,+) × Other [40,80)			0.0056	0.0057
			(0.0037)	(0.0037)
Own [80,+) × Other [80,+)			0.0125***	0.0126***
			(0.0043)	(0.0043)
Controls	Yes	Yes	Yes	Yes
Bureau controls	No	Yes	No	Yes
Category FE	Yes	Yes	Yes	Yes
Cycle-position FE	Yes	Yes	Yes	Yes
N	430,689	430,689	430,689	430,689
R^2	0.435	0.435	0.435	0.435

Notes. This table reports order-level regressions of the cash indicator on RappiCard utilization on the day before the order and the utilization of the household's other cards at the most recent credit bureau pull. Columns 1 and 2 enter both utilizations linearly with their interaction; columns 3 and 4 enter both in three bins, $[0, 40)$, $[40, 80)$, and $[80, +)$, with $[0, 40)$ omitted, and their interactions. All columns include user, bill-month, category, and cycle-position fixed effects, the controls of Table 2, and the log of the household's credit limits on other cards; columns 2 and 4 add the bureau controls of Table 3. The sample is orders with a credit bureau pull at most one month old, a positive limit on other cards, and other-card utilization at or below 110 percent, excluding orders with zero value or with day-before RappiCard utilization above 105 percent. Standard errors clustered by user in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D4: Substitution to payment instruments and utilization

	RappiCard		Other Card		Cash	
	(1)	(2)	(3)	(4)	(5)	(6)
Credit utilization, % (omitted bin [0,20))						
[20,40)	0.0146***	0.0124***	-0.0129***	-0.0105***	0.0006	0.0018
	(0.0025)	(0.0033)	(0.0023)	(0.0031)	(0.0010)	(0.0013)
[40,60)	0.0127***	0.0081**	-0.0137***	-0.0097***	0.0030***	0.0052***
	(0.0028)	(0.0036)	(0.0026)	(0.0034)	(0.0011)	(0.0015)
[60,80)	0.0028	0.0009	-0.0083***	-0.0067*	0.0050***	0.0071***
	(0.0031)	(0.0041)	(0.0027)	(0.0034)	(0.0012)	(0.0016)
[80,+)	-0.0851***	-0.0778***	0.0545***	0.0467***	0.0250***	0.0265***
	(0.0033)	(0.0042)	(0.0031)	(0.0040)	(0.0013)	(0.0017)
Revolving	-0.0348***	-0.0274***	0.0243***	0.0178***	0.0080***	0.0086***
	(0.0038)	(0.0047)	(0.0037)	(0.0044)	(0.0017)	(0.0021)
[20,40) × Revolving	-0.0065	-0.0062	0.0053	0.0053	-0.0005	-0.0024
	(0.0047)	(0.0057)	(0.0045)	(0.0053)	(0.0021)	(0.0026)
[40,60) × Revolving	-0.0185***	-0.0133**	0.0146***	0.0104*	0.0010	-0.0015
	(0.0049)	(0.0059)	(0.0047)	(0.0055)	(0.0022)	(0.0027)
[60,80) × Revolving	-0.0317***	-0.0217***	0.0255***	0.0177***	0.0066***	0.0029
	(0.0050)	(0.0063)	(0.0047)	(0.0057)	(0.0022)	(0.0028)
[80,+) × Revolving	-0.0543***	-0.0400***	0.0411***	0.0322***	0.0109***	0.0053**
	(0.0049)	(0.0061)	(0.0048)	(0.0058)	(0.0022)	(0.0027)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Bureau controls	No	Yes	No	Yes	No	Yes
Category FE	Yes	Yes	Yes	Yes	Yes	Yes
Cycle-position FE	Yes	Yes	Yes	Yes	Yes	Yes
N	1,255,790	810,411	1,255,790	810,411	1,255,790	810,411
R^2	0.440	0.473	0.421	0.453	0.398	0.416

Notes. This table reports estimates of equation (1) with three dependent variables: an indicator for paying with the RappiCard (columns 1 and 2), with another card (columns 3 and 4), and in cash (columns 5 and 6). The payment field does not distinguish credit from debit cards, so the other-card category contains both. Bins are defined on RappiCard utilization on the day before the order, with $[0, 20)$ omitted. Revolving equals one if the account carried an interest-bearing balance at the most recent statement before the order. The sample is the regression sample of Section 2, excluding orders with zero value or with day-before utilization above 105 percent. Even columns keep orders with credit bureau information. All columns include user, bill-month, category, and cycle-position fixed effects and the controls of Table 2; even columns add its bureau controls. Standard errors clustered by user in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D5: Bureau score and card utilization within household

	RappiCard utilization		Aggregate card utilization	
	(1)	(2)	(3)	(4)
Utilization, % (omitted bin [0,10))				
[10,20)	-0.45**		-1.23***	
	(0.20)		(0.22)	
[20,30)	-0.94***		-3.74***	
	(0.21)		(0.23)	
[30,40)	-3.25***		-7.85***	
	(0.22)		(0.24)	
[40,50)	-3.99***		-10.34***	
	(0.23)		(0.25)	
[50,60)	-5.12***		-12.51***	
	(0.23)		(0.27)	
[60,70)	-6.67***		-15.47***	
	(0.23)		(0.28)	
[70,80)	-7.50***		-17.27***	
	(0.24)		(0.29)	
[80,90)	-9.62***		-20.29***	
	(0.23)		(0.31)	
[90,105)	-12.13***		-23.95***	
	(0.21)		(0.32)	
Utilization (pp)		-0.129***		-0.269***
		(0.002)		(0.003)
Delinquent other card	-44.29***	-44.34***	-42.36***	-42.24***
	(0.48)	(0.48)	(0.48)	(0.48)
Delinquent RappiCard	-34.21***	-34.24***	-33.80***	-33.71***
	(0.66)	(0.66)	(0.65)	(0.65)
User FE	Yes	Yes	Yes	Yes
Pull-date FE	Yes	Yes	Yes	Yes
Mean BC score	648.3	648.3	648.3	648.3
Users	65,492	65,492	65,492	65,492
N (bureau pulls)	209,604	209,604	209,604	209,604
Within R^2	0.231	0.231	0.260	0.261

Notes. This table reports within-household regressions of the bureau (BC) score, in points, on card utilization. The unit of observation is a credit bureau pull. Columns 1 and 2 use RappiCard tradeline utilization in the bureau report; columns 3 and 4 use aggregate utilization across all of the household's open cards in the same report, including the RappiCard. Columns 1 and 3 enter utilization in ten-point bins with $[0, 10)$ omitted; columns 2 and 4 enter it linearly, in percentage points. All columns include indicators for a delinquent other card and a delinquent RappiCard, and user and pull-date fixed effects. The sample is pulls with both utilization measures below 105 percent. Standard errors clustered by user in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D6: Cash-credit gradient: neighborhood income and cash access

	Neighborhood income		Cash access (OXXO)	
	(1) Level	(2) Gradient	(3) Level	(4) Gradient
Income tercile T2	-0.0157***			
	(0.0052)			
Income tercile T3 (richest)	-0.0284***			
	(0.0036)			
Access tercile T2			0.0031	
			(0.0041)	
Access tercile T3 (farthest)			0.0070	
			(0.0044)	
Utilization [20,40)		-0.0026		0.0001
		(0.0025)		(0.0017)
Utilization [40,60)		0.0068**		0.0018
		(0.0027)		(0.0019)
Utilization [60,80)		0.0047*		0.0083***
		(0.0028)		(0.0020)
Utilization [80,+)		0.0398***		0.0305***
		(0.0030)		(0.0022)
Utilization [80,+) × T2		-0.0084**		0.0016
		(0.0039)		(0.0032)
Utilization [80,+) × T3		-0.0090**		0.0075**
		(0.0036)		(0.0035)
p: all bins × tercile		0.0003		0.0955
p: top bin × tercile		0.0311		0.0890
User FE	No	Yes	No	Yes
N	817,757	811,447	820,166	813,817
AGEB clusters	16,134	14,915	16,271	15,030
R^2	0.0212	0.3976	0.0197	0.3978

Notes. This table reports order-level regressions of the cash indicator on terciles of neighborhood income and of cash access, each in a level column and a gradient column. Neighborhood income is the CONAPO marginalization index of the household's AGEB, signed so that higher is richer; cash access is the distance from the household's home to the nearest OXXO convenience store, so T3 is farthest. Terciles are formed across households. The gradient columns interact the terciles with the utilization bins of equation (1), with $[0, 20)$ omitted; the p-values are from tests that all eight bin-by-tercile interactions are jointly zero and that the two top-bin interactions are zero. The sample is the regression sample of Section 2, excluding orders with zero value or day-before utilization above 105 percent, and restricted to households located in the AGEB corresponding to their app-launch coordinates. All columns include bill-month, category, and cycle-position fixed effects and the controls of Table 2; the gradient columns add user fixed effects. Standard errors clustered by AGEB in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D7: Cash-use gradient before and after the 70 percent utilization alert

	Indicator for cash payment	
	(1) Full sample	(2) Launch ±12m
Utilization [20,60)	0.0012	0.0014
	(0.0010)	(0.0014)
Utilization [60,70)	0.0043***	0.0050**
	(0.0016)	(0.0021)
Utilization [70,80)	0.0047***	0.0058***
	(0.0016)	(0.0021)
Utilization [80,+)	0.0324***	0.0298***
	(0.0015)	(0.0019)
Utilization [20,60) × Post	0.0015	0.0036
	(0.0016)	(0.0023)
Utilization [60,70) × Post	0.0046*	0.0082**
	(0.0025)	(0.0034)
Utilization [70,80) × Post	0.0087***	0.0100***
	(0.0024)	(0.0034)
Utilization [80,+) × Post	-0.0029	0.0037
	(0.0022)	(0.0029)
p: top bin × Post	0.1772	0.2064
N	1,255,790	698,866
R^2	0.398	0.401

Notes. This table reports order-level regressions of the cash indicator on utilization bins interacted with a post indicator for bill months from January 2023 on, when RappiCard began sending a push notification to every user with utilization above 70 percent. Bins are $[0, 20)$, $[20, 60)$, $[60, 70)$, $[70, 80)$, and $[80, +)$, with $[0, 20)$ omitted; the bill-month fixed effects absorb the post main effect. Column 1 uses the full sample and column 2 the 12 bill months before and after the launch. The p-value is from a test that the top-bin interaction is zero. The sample is the regression sample of Section 2, excluding orders with zero value or with day-before utilization above 105 percent. All columns include user, bill-month, category, and cycle-position fixed effects and the controls of Table 2. Standard errors clustered by user in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D8: Cash use and temptation goods

	Indicator for cash payment	
	(1)	(2)
Temptation good	0.0003	0.0003
	(0.0009)	(0.0009)
Revolving	0.0200***	0.0169***
	(0.0010)	(0.0010)
Temptation good × Revolving	-0.0078***	-0.0077***
	(0.0015)	(0.0015)
Credit utilization		0.0311***
		(0.0011)
Log order value		0.0007*
		(0.0004)
Any purchase last month		0.0015
		(0.0031)
Log purchase last month		-0.0018***
		(0.0004)
Controls	No	Yes
Cycle-position FE	No	Yes
N	1,462,258	1,432,502
R^2	0.388	0.390

Notes. This table reports order-level regressions of the cash indicator on a temptation-good indicator, a revolving indicator, and their interaction. Temptation goods are orders containing alcohol or tobacco. Revolving equals one if the account carried an interest-bearing balance at the most recent statement before the order. Column 2 adds cycle-position fixed effects, the controls of Table 2, and day-before utilization entered linearly. The sample is the regression sample of Section 2, excluding orders with zero value. Both columns include user and bill-month fixed effects. Standard errors clustered by user in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D9: Covariate balance at the cutoff

Covariate	Jump in SD	Honest 95% CI (SD)	Eff. N
Pre-period Rappi and payment behavior (90 days before the campaign)			
Cash share	-0.000	[-0.111, +0.110]	6,816
Cash-paid order count	-0.053	[-0.162, +0.055]	5,129
Cash-while-revolving share	+0.007	[-0.129, +0.144]	4,703
Rappi order count	+0.023	[-0.088, +0.134]	4,154
Rappi order value*	+0.002	[-0.121, +0.124]	3,818
External-card share	+0.010	[-0.108, +0.129]	5,924
Account state at campaign evaluation			
Prior CLI count	-0.026	[-0.147, +0.094]	4,842
Utilization at evaluation	+0.014	[-0.101, +0.129]	5,835
Utilization (3m avg.)	-0.004	[-0.120, +0.112]	5,446
Balance*	+0.048	[-0.071, +0.167]	5,037
Credit line*	+0.032	[-0.052, +0.117]	5,118
Total payments*	-0.009	[-0.102, +0.084]	6,782
Payments / total balance (3m)	-0.039	[-0.136, +0.059]	13,657
Internal risk score (RC)	-0.011	[-0.099, +0.077]	6,860
Internal collection score (ICC)	+0.071	[-0.054, +0.196]	4,502
Tenure (months)	+0.060	[-0.039, +0.159]	8,051
Months since last CLI	+0.077	[-0.045, +0.198]	5,178
Payment and leverage at last closed statement			
Revolving	+0.051	[-0.073, +0.175]	5,589
Revolving balance (interest-bearing)*	+0.094	[-0.041, +0.229]	4,735
Installment balance (interest-free MSI)*	-0.048	[-0.145, +0.049]	7,591
Purchases*	-0.026	[-0.174, +0.122]	2,347
Paid on time*	+0.064	[-0.103, +0.231]	2,349
Credit bureau			
Total debt‡	+0.017	[-0.149, +0.183]	1,438
Total credit limit, all cards‡	-0.006	[-0.170, +0.158]	1,071
Bureau tenure (months)‡	+0.050	[-0.149, +0.249]	1,898
Holds other cards‡	+0.064	[-0.111, +0.238]	2,820
Demographics			
Age	+0.041	[-0.101, +0.183]	3,750
Male	-0.001	[-0.091, +0.089]	9,712

Notes. This table reports the change at the credit score cutoff in each predetermined covariate, standardized by the covariate's standard deviation, on the sample of Table 5. Estimates are by RDHonest with a triangular kernel and the bandwidth chosen by the estimator; effective N is the number of observations within it. Pre-period Rappi and payment variables are measured over the 90 days before the campaign. * marks peso amounts winsorized at 2 and 98 percent; ‡ marks credit bureau variables, available for a subsample. Two covariates with no variation at the cutoff, the count of unpaid statements and an indicator for falling into non-payment, are omitted. Brackets are honest 95 percent confidence intervals from standard errors clustered by user.

Table D10: Robustness: triangular and uniform kernels

	Triangular kernel			Uniform kernel		
	Jump	95% CI	h	Jump	95% CI	h
Panel A. First stage: accepted the increase						
All	$+0.185^{**}$	$[+0.152, +0.219]$	12.1	$+0.172^{**}$	$[+0.141, +0.204]$	6.0
Utilization < 0.6	$+0.300^{**}$	$[+0.253, +0.347]$	15.1	$+0.302^{**}$	$[+0.253, +0.351]$	11.0
Utilization ≥ 0.6	$+0.123^{**}$	$[+0.059, +0.187]$	16.0	$+0.099^{**}$	$[+0.063, +0.136]$	5.0
Difference	-0.177	$p < 0.01$		-0.203	$p < 0.01$	
Panel B. Cash-paid orders						
All	-0.133^{**}	$[-0.247, -0.019]$	12.1	-0.124^{**}	$[-0.236, -0.013]$	8.0
Utilization < 0.6	-0.058	$[-0.243, +0.127]$	15.1	-0.127	$[-0.304, +0.051]$	12.0
Utilization ≥ 0.6	-0.140^{**}	$[-0.261, -0.019]$	16.0	-0.127^{**}	$[-0.246, -0.009]$	12.0
Difference	-0.082	$p = 0.39$		-0.001	$p = 0.99$	
Panel C. asinh*(cash-paid orders)*						
All	-0.040^{*}	$[-0.087, +0.007]$	10.1	-0.040	$[-0.088, +0.008]$	7.0
Utilization < 0.6	$+0.000$	$[-0.067, +0.067]$	11.6	$+0.007$	$[-0.062, +0.075]$	8.0
Utilization ≥ 0.6	-0.066^{**}	$[-0.119, -0.013]$	13.9	-0.080^{**}	$[-0.134, -0.027]$	10.0
Difference	-0.066	$p = 0.10$		-0.087	$p = 0.04$	
Panel D. $\mathbf{1}\{$*any cash-paid order*$\}$						
All	-0.018	$[-0.052, +0.016]$	9.2	-0.021	$[-0.056, +0.014]$	7.0
Utilization < 0.6	$+0.020$	$[-0.027, +0.067]$	10.9	$+0.022$	$[-0.027, +0.071]$	8.0
Utilization ≥ 0.6	-0.045^{**}	$[-0.083, -0.008]$	13.8	-0.056^{**}	$[-0.094, -0.018]$	10.0
Difference	-0.066	$p = 0.02$		-0.078	$p < 0.01$	

Notes. This table reports the estimates of Table 5 (Panel A) and Figure 7 (Panels B to D) under the triangular kernel and under a uniform kernel. The triangular columns repeat those estimates; under the uniform kernel, M and the bandwidth h, in score points, are re-selected by the estimator. First-stage jumps are shares. All estimates include campaign fixed effects. The difference rows report the high-utilization minus the low-utilization jump, with the p-value of a test of equal jumps that treats the two cells as independent. Brackets are honest 95 percent confidence intervals from standard errors clustered by user. ** marks an honest 95 percent interval that excludes zero and * an honest 90 percent interval that excludes zero.

Table D11: Spending and repayment response to CLI

	RD jump, scaled by the old limit		Per peso of limit granted	
	Estimate	95% CI	Estimate	95% CI
Limit granted, $\Delta L/L_{\text{old}}$	+0.0968	[+0.0803, +0.1133]	–	
Card spending, days 1–30	+0.0055	[−0.0308, +0.0418]	+0.057	[−0.28, +0.40]
days 1–60	+0.0173	[−0.0519, +0.0865]	+0.179	[−0.47, +0.83]
days 1–90	+0.0223	[−0.0826, +0.1272]	+0.230	[−0.76, +1.22]
Repayment, days 1–30	−0.0118	[−0.0473, +0.0236]	−0.122	[−0.46, +0.21]
days 1–60	+0.0134	[−0.0501, +0.0770]	+0.139	[−0.46, +0.74]
days 1–90	+0.0185	[−0.0820, +0.1189]	+0.191	[−0.76, +1.14]

Notes. This table reports the change at the credit score cutoff in the limit granted, in RappiCard spending, and in repayment, each cumulated over the window and divided by the pre-campaign credit limit, pooled across utilization and revolving status, on the sample of Table 5. Estimates are by RDHonest with campaign fixed effects and a triangular kernel, with the bandwidth chosen by the estimator; the spending and repayment estimates also control for the pre-campaign level of the same outcome. The last two columns divide each spending or repayment jump by the jump in the limit granted, with a delta-method 95 percent interval. Brackets in the first two columns are honest 95 percent confidence intervals from standard errors clustered by user.

Table D12: Cash use and repayment release of utilization

	Indicator for cash payment			
	(1)	(2)	(3)	(4)
Δu (utilization released)	-0.039***	-0.033***	-0.022***	-0.031***
	(0.003)	(0.005)	(0.007)	(0.006)
u (utilization before the repayment)	0.023***	0.029***	0.025***	0.021***
	(0.002)	(0.002)	(0.003)	(0.002)
$\Delta u \times u$		-0.015***	-0.016**	-0.004
		(0.006)	(0.008)	(0.006)
Revolving $\times \Delta u$	-0.023***			0.004
	(0.005)			(0.013)
Revolving $\times \Delta u \times u$				-0.039***
				(0.013)
Controls	No	Yes	Yes	Yes
Bureau controls	No	No	Yes	No
N	633,750	626,105	324,481	626,105
R^2	0.407	0.408	0.435	0.408

Notes. This table reports estimates from regressions of an indicator for cash payment on the utilization released by a RappiCard repayment, on orders placed within ten days after the repayment. Δu is the fall in utilization from the last balance event before the repayment posts to the first event on or after it, capped at one. u is utilization at the last balance event before the repayment posts, winsorized at the 2nd and 98th percentiles. Revolving equals one if the account carried an interest-bearing balance at the most recent statement before the order. Column 1 interacts Δu with the revolving indicator, columns 2 and 3 interact it with u, and column 4 adds the triple interaction. The revolving main effect in columns 1 and 4 and its interaction with u in column 4 are included but not reported. The sample is the regression sample of Section 2, excluding orders with zero value or with day-before utilization above 105 percent, and restricted to orders with a matched repayment in the ten days before the order. Column 3 keeps orders with credit bureau information and a positive credit limit on other cards. All columns include the days since the repayment and its square, and user, bill-month, and category fixed effects. Controls are the log of order value, the log of RappiCard purchases in the previous billing cycle, and an indicator for no purchases in that cycle. Bureau controls are an indicator for any other delinquent tradeline and utilization on the household's other credit cards, capped at one. Standard errors clustered by user in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D13: Robustness of the expected overflow price

			Shadow cost, non-revolving	
	$\mathbb{E}[\theta]\rho$	95 percent interval	20%	80%
Headline (Weibull F)	0.434	[0.312, 0.510]	4.8	24.4
Distribution of the cash cost				
Lognormal F	0.425		4.7	23.9
Log-logistic F	0.433		4.8	24.3
Shock size				
Mean shock 1,500 pesos	0.437		1.2	15.4
Mean shock 6,000 pesos	0.397		10.0	30.1
Rises with the total credit limit (elasticity 0.14)	0.438		4.7	24.7
Households fitted separately				
RappiCard at least 70% of credit card limits	0.419	[0.249, 0.540]	6.7	25.7
Other households	0.440	[0.278, 0.547]	4.1	23.6
Interest inputs				
Utilization measured over the whole panel	0.357	[0.242, 0.440]	3.9	20.2

Notes. This table reports $\mathbb{E}[\theta]\rho = (p\,\theta_m + q)\,\rho$, the expected overflow price per dollar of expenditure shock for a household with no available credit, estimated as in Table 7 with one change per row. Intervals are from 500 bootstrap draws (Appendix B.2); the rows without an interval are point estimates. The last two columns report the shadow cost of a non-revolving household at 20 and 80 percent utilization, in cents per dollar, as point estimates. In the mean-shock rows, the cap of G is ten times the mean and the measured hold amount is unchanged. The two groups of households are disjoint: 19,990 households for which RappiCard holds at least 70 percent of their credit card limits at card approval, and 39,921 other households, each fitted to its own ten cash shares. In the last row, the utilization bins of $\kappa^R(u)$ and of the probability that a non-revolving cycle turns revolving are measured over the whole panel instead of the first 12 months after card approval; that probability is held fixed in the bootstrap of this row.